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Javon Hickmon

Moral Images and (Im)moral Machines:
Toward Cross-Modal VLM Value Consistency

Javon Hickmon

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Ali Farhadi

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Abstract

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Javon Hickmon

Chair of the Supervisory Committee:
Ali Farhadi
Department of Computer Science & Engineering

Computer science has made significant progress in developing machine learning systems that produce outputs aligned with human ethical expectations. Despite these advances, much of the literature on AI ethics engages only superficially with the philosophical foundations of moral value and the implications these foundations have for the design and evaluation of intelligent systems. This gap is particularly relevant in multimodal machine learning. As vision-language models (VLMs) become increasingly integrated into real-world applications, ensuring that they can make reliable moral and value-based judgments has emerged as an important challenge.

This thesis addresses this problem through two complementary contributions. Chapters 1–3 develop the philosophy by examining the nature of intrinsic value and its relevance to human-computer interaction and multimodal machine learning. These chapters aim to establish a conceptual framework for understanding value as a meaningful attribute to evaluate. Chapter 4 builds upon this foundation by empirically investigating the cross-modal consistency of moral judgments in VLMs. This work demonstrates that many widely used VLMs fail to maintain consistent value representations across modalities, suggesting that current multimodal architectures do not reliably encode value-laden concepts within the image-text embedding space. This calls into question the validity of downstream applications and evaluations that assume stable moral reasoning across modalities.

TABLE OF CONTENTS

	Page
List of Figures	iii
Chapter 1: Intrinsic Value in Human-Computer Interaction	1
1.1 Introduction	1
1.2 The Intrinsic Value of Human-Human Interaction	2
1.3 The Intrinsic Value of Human-Computer Interaction	9
1.4 The Inherent Value of Cultural Diversity	13
Chapter 2: In Defense of Moral Realism: The Power of Rational Reflection	20
2.1 Introduction	20
2.2 Rawls’ Reflective Equilibrium	23
2.3 The Limits of Rational Reflection	25
2.4 An Answer to the Possibility of Influence	32
2.5 An Answer to the Independence of Future Judgments	33
2.6 What About the Darwinian Dilemma?	34
Chapter 3: Moral Influence	35
3.1 Introduction	35
3.2 The Intrinsic Moral Value of Media	35
3.3 From Moral Influence to Moral Reactions	38
3.4 From Moral Reactions to Moral Judgments	42
Chapter 4: The Cross-Modal Consistency of VLM Value Judgments	45
4.1 Introduction	45
4.2 Related Work	46
4.3 Methods	48
4.4 Results	59
Bibliography	75

Appendix A: All imSitu Figures	79
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LIST OF FIGURES

Figure Number		Page
3.1	The Terror of War (also known as “Napalm Girl”), June 8, 1972: Kim Phúc, centre, running down a road after a South Vietnam Air Force napalm attack [36]	39
4.1	The (non-normalized) cross-modal modal probabilities of generating the token “moral” from LLaVa-7B given 10,000 random samples from imSitu . . .	59
4.2	Plots of the top 10 tokens for LLaVa-7B when conditioned upon the specified scenario as part of the prompt with value pair being moral/immoral (See Section 4.3.4 for structure)	60
4.3	The cross-modal modal normalized positive value probability for the value pair moral/immoral from LLaVa-7B on imSitu	61
4.4	Plots of the cross-modal normalized positive value probabilities of LLaVa-7B with the specified value pairs.	62
4.5	Plots of the cross-modal normalized positive value probability of the specified models on the value pair moral/immoral.	64
4.6	A heatmap representing the probability of generating the target text given the defined image using LLaVa-7B. The ground truth falls along the diagonal.	66
4.7	The cross-modal normalized positive value probability of LLaVa-7B for the token pair moral/immoral. Incorrectly classified samples are colored orange. .	67
4.8	Plots of the image classification results in the cross-modal normalized positive value probability of LLaVa-7B. Each subplot uses the specified value pair. . .	68
4.9	Plots of the image classification results in the cross-modal normalized positive value probability of LLaVa-7B. Each subplot uses the specified value pair. . .	70
4.10	Cross-modal normalized positive value probability plots in order of increasing scale for each model, using the value pair <i>moral/immoral</i>	73
A.1	LLaVa-7B cross-modal normalized positive value probabilities on imSitu . . .	79
A.2	LLaVa-13B cross-modal normalized positive value probabilities on imSitu . .	79
A.3	InternVL3-1B cross-modal normalized positive value probabilities on imSitu .	80
A.4	InternVL3-2B cross-modal normalized positive value probabilities on imSitu .	80
A.5	InternVL3-8B cross-modal normalized positive value probabilities on imSitu .	80
A.6	Qwen3VL-2B cross-modal normalized positive value probabilities on imSitu .	81

A.7	Qwen3VL-4B cross-modal normalized positive value probabilities on imSitu .	81
A.8	Qwen3VL-8B cross-modal normalized positive value probabilities on imSitu .	81
A.9	InstructBLIP-7B cross-modal normalized positive value probabilities on imSitu	82
A.10	InstructBLIP-13B cross-modal normalized positive value probabilities on im-Situ	82
A.11	Molmo-7B cross-modal normalized positive value probabilities on imSitu . . .	82

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DEDICATION

to my mom, and sister

Chapter 1

INTRINSIC VALUE IN HUMAN-COMPUTER INTERACTION

1.1 Introduction

The concept of *value* is particularly evasive because it can take on different meanings depending on context. Outside of philosophy, value can be used to mean a variety of moral and non-moral ideas. Excellence in art and music can be considered valuable, yet scientific accomplishments are also considered valuable, although it is generally accepted that these are valuable in different respects. The term *value* is also frequently exchanged with *moral* or *good*, but when we look closer at these terms, their distinction becomes apparent. Value, and specifically value theory within philosophy, typically encompasses whatever areas are deemed to encompass that which can be evaluated or has some otherwise evaluative aspect.

This becomes complex when we begin to explore which things *have* value or are valuable. W. D. Ross showcases the nuance that is often difficult to capture in definitions such as this: “For most theories of value may be divided into those which treat it as a quality and those which treat it as a relation between that which has value and something else...” [31, p. 75]. He later clarifies the concept of value to include *intrinsic value*. The form of intrinsic value described is objective universal value, or value that would exist regardless of whether there was an observer. Intrinsic value takes the core concept of value a step further by defining value to be an *attribute* of certain things. Intrinsic value has a direct correspondence to morality or “rightness.” “Rightness no more than beauty is an intrinsic value, and that the only correct way of expressing intrinsic value is the word ‘good’” [31, p. 133].

This chapter will focus on sharpening the distinct human values that affect our understanding of machine ethics, specifically the ways that value becomes present and influenced by human-computer interaction (HCI). The resulting distinctions created from this exploration of human value set the foundation of my work exploring value representation within multimodal models. Three values in particular are relevant to this discussion: the value of

human-human interaction (also called interpersonal value), the value of human-computer interaction, and the value of cultural diversity (which informs epistemic value). Each of these fundamentally addresses different questions regarding the role of people in parts of the machine learning process, questions that cannot be answered in a single work but will be discussed in depth as a means of detailing future work to be conducted within the field.

1.2 The Intrinsic Value of Human-Human Interaction

Within this discussion, where I examine the value of human-human (pronounced “human to human”) interaction, I will be intentionally not arguing for any particular value theory, but rather, I will argue against the set of theories that reject the intrinsic value of actions. I reject this view because it has significant negative implications regarding responsibility and, by proxy, the prioritization of values.

1.2.1 The Intrinsic Value of Actions

With our earlier definition of value being that which encompasses some evaluative aspect, in particular with relation to “the good,” we can clearly see that value could either be positive or negative. In this case, positive value contributes to “the good” while negative value contributes to “the bad.” This means that certain things can be neutral concerning value, neither contributing to “the good” nor “the bad.” In addition, I must clarify what is meant by “intrinsic” value. I define intrinsic to mean that which is core to the nature of being of something. In other words, a thing and its intrinsic value are inextricable. Finally, I define an “action” to be the process of doing something.

These definitions are fundamental to the way that I will defend my stance on the intrinsic value of actions. When we think about an action as being the process of doing something, it becomes clear that all actions can have an associated state. An action need not have an associated state; for example, the action “jump” has no state being described. Despite this, the state of an action is not separate from the action itself, so the act of “jumping on the grass” is not distinct from “jumping,” but rather it is a subset of, and gives context to, the central action.

With this in mind, I argue that actions *can* have value. They do not need to have value, but certain actions are directly related to “the good” or “the bad”. For example, I assert that “jailing the innocent” is an action that is bad on its own or has intrinsic negative value. This assertion that actions in and of themselves can have value directly implies that the value of an action can be completely separate from one’s motivation to complete an action, or even the consequences of the action. I can assert that “jailing the innocent” is intrinsically wrong, no matter if there was a misunderstanding and someone genuinely thought the innocent person was guilty, or if the innocent person was jailed to reduce some political unrest.

This distinction of the intent and consequences being irrelevant is core to our definition of “intrinsic,” because if these factors were to modify the evaluation, then the value would not be core to the action in and of itself. G. E. Moore’s conception of value makes clear how we can understand this distinction for ourselves: “In order to arrive at a correct decision on the first part of this question [‘What things have intrinsic value, and in what degrees’], it is necessary to consider what things are such that, if they existed by themselves, in absolute isolation, we should yet judge their existence to be good; and, in order to decide upon the relative degrees of value of different things, we must similarly consider what comparative value seems to attach to the isolated existence of each” [27, p. 187]. Effectively, Moore states that the evaluation of a thing in isolation makes it possible to identify whether it is good that it occurs. This distinction will become extremely important in our later discussion on the teleological view of value.

Here, it is important to comment on our understanding of the inextricable nature of state and action. This core nature means that state *can* augment the value of an action as well. “Punishing the innocent” may have positive value, but “punishing the guilty” may have negative value. The core action of “punishing” remains constant, but the state (or context) of the action affects the evaluation. Finally, core to this section is the connection between actions and interactions. Interaction is defined as a mutual action of two things upon one another.

1.2.2 *The Intrinsic Value of Human-Human Interaction*

We have already established that some actions, and by relation, some interactions can have intrinsic value. Now we are faced with answering the question surrounding which types of interactions can have intrinsic value. To start this discussion, we will examine human-human interaction through a thought experiment:

David needs to run a few errands before going home. He stops at the bank to withdraw some money for his train ticket. He says hi to the teller, withdraws his money, and then heads to the train station. When he gets to the station, he walks up to the ticket salesperson, pays for his ticket, and then boards the train. Upon arriving at home, he realizes he's quite hungry, so he orders delivery and waits for the driver to get there. After 15 minutes, his phone alerts him that his order has been delivered, so he goes outside to get the food from the driver. After getting the food, he has dinner and then goes to sleep for the night.

In each of these interactions, David shared some form of interpersonal interaction; however, he had no preexisting relationships with any of the people. This is an important characteristic of these scenarios. Many would quickly label interactions with friends as valuable in some way, but interactions with strangers seem to elicit a different response, or at least it appears to be valuable in different or lesser ways than friendship is valuable. To further understand this nuance, let's push the thought experiment to its extremes:

David needs to run a few errands before going home. He stops at the bank to withdraw some money for his train ticket and is surprised to find no human tellers inside. Instead, there are only ATMs in the bank. Despite this, he quickly gets the money and heads to the train station. When he gets to the station, again, he is confronted with a ticket machine. He pays for his ticket and then boards the train. Upon arriving at home, he realizes he's quite hungry, so he orders delivery and waits for the driver to get there. After 15 minutes, his phone alerts him that his order has been delivered, so he goes outside to find a self-driving vehicle with his food. After getting the food, he has dinner and then goes to sleep for the night.

With this revised version of the thought experiment, one major question arises: can human-human interaction have intrinsic value?

The Teleological View of Value

It is important to note that the aforementioned question asks whether human-human interaction is more valuable than no interaction at all. Thinking back to G.E. Moore’s framing of intrinsic value, it becomes clear why there are multiple ways that philosophers have understood ways to evaluate a concept for its goodness. One such view is the teleological perspective:

“The primary bearers of value are states of affairs, or, over time, ways the world might go. These things can have intrinsic value, that is to say, value that is not a matter of their tendency to contribute to or make possible something else which is of value” [32, p. 79]. Upon this view of value, the final state of affairs (in other words, “the ends”) are the only things that can have value. This can be formulated to fit Moore’s earlier formulation of value, where we as agents evaluate whether the state of the world with this thing in isolation is better than the state of the world without, and therefore whether this thing should be promoted. This is an important aspect of the teleological standpoint, where actions or interactions are intrinsically valuable insofar as they bring about a state of affairs. “To be (intrinsically) valuable, on such a view, is to be ‘to be promoted’” [32, p. 80].

To connect this understanding back to the thought experiment, under the teleological view of value, David’s human-human interaction *was* intrinsically valuable; however, the interaction was *only* valuable insofar as it helped him complete his errands. This distinction is necessary because the view holds that human interaction *on its own* was not intrinsically valuable. Now, one could certainly formulate the thought experiment in a way such that the interpersonal interaction has some sort of increase in “the good” of the world. Maybe David’s exchanging pleasantries with the bank teller resulted in increased dopamine, which improved David’s mood, resulting in a better state of the world. This would be completely reasonable under the teleological perspective, but it still shows that the human interaction was only valuable because it brought about a better state.

The teleological view of value also addresses the ways that we, as agents, ought to be motivated by value. “As agents, our relation to states of affairs lies in being able to realize them, to prevent them from occurring, or to make their occurrence more or less likely. What we have reason to do, on this view (at least as far as questions of value are concerned), is to act to realize those states of affairs that are best - that is, have the greatest value” [32, p. 80]. Therefore, the teleological perspective asserts that an agent *must* be primarily motivated by the improved state of affairs.

Rejecting the Teleological View of Value

Using this refined understanding of the teleological definition of intrinsic value, we may assert interpersonal interaction as valuable if the states of affairs resulting from it are “to be promoted.” This assertion seems innocuous at first, and many would agree to it at first glance. The states of affairs resulting from interpersonal interaction are often good things that one would want to be promoted. For instance, interpersonal interaction is the foundation for scientific collaboration. Scientific collaboration allows for the sharing of knowledge, thereby creating a process where new knowledge can be discovered, which is something that ought to be promoted. Despite these points in favor of the teleological understanding of value, it hides one major implication: the relationship between our reasons to promote something and the state of affairs that it achieves.

To reiterate, the teleological view states that interactions can only be valuable based on the state of affairs that are brought about. This means that for one to have reason to pursue interpersonal interaction, they must **first** identify some way in which the world would be better upon a modified state of affairs, **then** explain how the state of affairs resulting from the interaction works toward this better state of the world. This poses a significant problem and is the reason why I reject the teleological view. My rationale is as follows: even if human-human interaction is aimed at some end, it doesn’t necessarily mean that we pursue such interaction simply *because* we are focused on the ends. This rejection of the teleological view of value shares a similar form to T. M. Scanlon’s objection to the same view, citing one’s reasons for pursuing scientific study as a major counterargument

against the teleological perspective. “In particular, even though the actions involved in scientific inquiry and study are each aimed at some end, the best account of our reasons for those actions may not flow *from* the value of these results *to* our concern with them” [32, p. 93]. Note that I am not saying values cannot take a teleological form; rather, I simply argue that some types of value are not teleological in form, human-human interaction being a prominent example.

From this, there are many implications because it shows that the final state of affairs is not the only thing that can hold inherent value; rather, the human-human interactions themselves can be inherently valuable on their own merit, and not necessarily as a means to achieve some better state of the world. For instance, even if David’s greeting of the bank teller resulted in an increase of dopamine, it is unlikely that this was his primary motivation in greeting them. David could have just as easily felt no “reward signal” from greeting the teller, and yet could still be motivated to do such a thing. An alternative, and I think more likely, account was that David understood there to be a good reason to be kind to other people. Greeting people you interact with is one way to be kind to them; therefore, David was motivated to greet the bank teller. Notice how this account starts with one’s motivations for doing something, *then* flows to the value of the results.

Again, one could object to this view, stating that David was unintentionally motivated by the dopamine reward signal to view being kind to other people as good, but this objection could easily be defeated by examining a case where David would have no external motivation to be kind to others. We could imagine the scenario where every time David greets someone, he is publicly ridiculed; however, David has some internal motivation to continue being kind to others. For instance, perhaps he is a Christian and abides by the golden rule, “whatever you wish that others would do to you, do also to them” ([1] Matthew 7:12), or perhaps he has some other motivation to be an uplifting person; no matter whether you agree with his motivation or not, it becomes apparent that this would mean that his motivation is completely removed from the state of affairs.

When is a Value Valuable?

One may note that this example has slightly diverged from our primary line of discussion up until now. In our analysis of the teleological view of value, we focused on ways to distinguish which things are inherently valuable; however, these recent examples have focused on the values that one can hold. Although these are distinct questions, they are related. I do not mean to say that every value people hold is a good one; on the contrary, it is quite common for people to be mistaken in the values that they hold. Even without these flaws in how values are held, I argue that there is a fundamental difference between the values that one holds and what is deemed to be valuable. For example, one might argue that they value their family, they love their parents and siblings, and connecting with them is something that is good, or at least can be aligned with “the good.” Despite this value, it would seem odd or misplaced to argue that their family is valuable.

When we assert something to be valuable, we mean that it holds value (or should hold value) for all people in all places for all time. Those who do not value what is valuable are missing something or otherwise misaligned with “the good.” In fact, being valuable directly corresponds to something greater, where it implies a direct relation to “the good” as an observer-independent trait. G.E. Moore’s conception of value showcases this observer independence, where, upon isolation, a thing is evaluated to be good. This alignment with “the good” holds implications for normative ethical theory, where individuals *ought* to pursue the good. Even if a theory does not hold to the strict normativity of value, at the very least, alignment with “the good” constitutes a *pro tanto* reason for pursuing that which is valuable. If all people of all times and all places should pursue “the good,” then they should pursue what is valuable. This exposes the core issue regarding why one might value their family without claiming them to be valuable.

With this clarified understanding of value and that which is valuable, it becomes easier to understand why certain values are and aren’t valuable. It can be easy to fall into the teleological trap of assuming things are only valuable insofar as their outcomes; however, we have seen that this produces counterintuitive outcomes. In this sense, I argue for a form of T. M. Scanlon’s contractualism, where the connection between the values people

hold and what is valuable comes down to the reasons one has for holding such values. This idea of reasonable rejection brings up more questions regarding what/who determines what is reasonable and what makes a rejection reasonable, all questions that are outside of the scope of this thesis. Despite this, Scanlon’s theory is enough to clearly show how values are connected to that which is valuable. To claim that interpersonal interaction is valuable amounts to claiming that there are *good* reasons to value interpersonal interaction. This is an important distinction because it exposes the fact that there are good reasons to value interpersonal interaction, and these reasons need not be teleological in structure. Once again, I am not arguing for a specific value theory within this thesis; rather, I aim to utilize this clear formulation of value to show the ways that the fundamental philosophical conception of value connects to modern works concerning ethics within computer science.

1.3 The Intrinsic Value of Human-Computer Interaction

Despite this intrinsic value of human-human interaction, the thought experiment also elicits the question: “Can human-computer interaction have intrinsic value?” From the ATM, to the ticket machine, to the self-driving delivery vehicle, each of these were interactions with computers. This question is fundamentally more divisive than our exploration of human-human interaction. As a result, I will use this moment to re-clarify our definition of “value” and our definition of “intrinsic.”

Value often implicitly means “that which is good,” or more broadly, that which corresponds to “the good.” Despite this implicit understanding, the core definition of value is “that which has some evaluative aspect,” meaning value could correspond to either “the good” or “the bad.” Therefore, a better way to frame our understanding of value is in terms of positive and negative value to disambiguate the two. In a similar vein, the word *intrinsic* (as it relates to value) directly corresponds to G.E. Moore’s conception of value, where the goodness of a thing must be considered in isolation and not in relation to any motivation or outcome.

With these definitions in mind, the initial response of many to the question regarding the intrinsic value of human-computer interaction would be “it depends.” This is because many are aware that context can affect the way that an interaction is evaluated. For

example, creating a button so that users can cancel their subscription would be a good thing, but making the button difficult to find would be a bad thing. Still, this evaluation of “it depends” implies that interacting with computers *can* be valuable, whether it be your phone allowing you to connect with friends, or a calculator calculating a particularly tricky set of mathematical operations.

1.3.1 *The Problem of Mediation*

A potential counterargument to this view would hold that interactions are only as valuable as the things they mediate. For example, interacting with friends via text might be valuable, but the core technology of text messaging is not inherently valuable. This argument becomes less straightforward upon examination of the second example of calculators. Calculators leverage the core design of computers (to compute, i.e., determine a result through mathematical/logical means); therefore, we can formulate a similar question by asking whether mathematics, or rather, the application of mathematics, is inherently valuable. I would argue yes, in the same way that the application of scientific knowledge is inherently valuable, but one might note that this is a slightly different question. The two examples just described emphasize different interactions, where texting friends showcases an interaction that uses a computer, while using a calculator focuses on the interaction with the computer itself. In other words, the former describes interaction mediated *through* a computer, while the latter describes interaction *with* a computer.

This distinction helps to clarify our core question: “Can human-computer interaction have intrinsic value?” Our focus is on interacting *with* a computer rather than interacting *through* a computer. This helps to address the counterargument because it shows how the interaction that occurs through a computer is separate from the interaction that occurs with the computer. Note that interacting through a computer *requires* interaction with a computer, but these are evaluated separately. For example, the value of texting a friend is not necessarily connected to the value of texting, in the same way that the value of texting an enemy is not necessarily connected to the value of texting. Rather, in these scenarios, texting serves as a way for individuals to interact, but the technology of text messaging

could be evaluated on its own. Therefore, the value of text messaging may facilitate other scenarios, but it is distinct and can be evaluated on its own.

A second counterargument might point out how, similar to human-human interaction, human-computer interaction is still dependent on context, where computing a new scientific discovery would carry intrinsic positive value, but computing and hacking private information would have negative value. To defend against this, I simply reiterate the definition of an action. Put simply, an action is something that an agent does. In other words, it is the process of doing something. This idea of “process” means that there can be a state associated with an action. This state can augment the value of an action, where punishing a murderer might be evaluated as good or as contributing positive value, but punishing an innocent person ought to be evaluated as bad or as contributing negative value. With our understanding that interaction refers to action, which refers to two or more things acting upon one another, it becomes clear that the state associated with an interaction *is* the context mentioned within the counterargument. Therefore, the state of an interaction is core to its intrinsic value, allowing for positive or negative situations to potentially alter its evaluation.

1.3.2 *The Invaluability of Human-Human Interaction*

Close readers may have noted that the prior thought experiment included another aspect that has not yet been discussed. By providing the example of David having human interaction during his errands, *in comparison to* the example of him not having these interactions, a second question was subtly asked: “Can human-human interaction be more valuable than human-computer interaction?” To assess this comparison, we once again look to G. E. Moore’s formulation of value, specifically when he writes, “to decide upon the relative degrees of value of different things, we must similarly consider what comparative value seems to attach to the isolated existence of each” [27, p. 187]. To assess the relative value of human-human interaction in comparison to that of human-computer interaction, we must explore both in isolation, then evaluate whether the intrinsic value of one captures more of “the good” than the other.

In the context of our thought experiment, David's interactions with humans and computers brought about the same outcome; however, our assessment of value rejected the teleological view. Thus, if one were to envision a world where the only thing that existed were David's interactions, would interacting with people or interacting with solely computers be best? This question is much easier to answer; interacting with people would be more valuable than never interacting with people. Although this example seems extreme, it exposes where our priorities lie when decisions must be made with regard to valuing people over productivity. In other words, **human-human interaction is invaluable**. Invaluability means that a thing ought not to disappear, otherwise something would be "wrong" with the world. Once again, there is nuance to this. The removal of *all* human-human interaction would undoubtedly result in negative value, but the removal of *some* human-human interaction could result in positive value.

For example, the removal of human telephone operators allowed for better scaling so that more people could use and operate telephones on their own. This was difficult because hundreds of thousands of jobs, and likely millions of human-human interactions, were lost due to this shift. It can be very easy to view this scenario through the teleological perspective, stating that the outcome brought about more value than the initial state of things, so it was valuable; however, I would argue for a different perspective on this scenario. The human-human interaction via telephone operators *was* valuable, and its removal brought about negative value, even though automatic dialing brought about many other technological innovations. This inherent negative value of removing human-human interactions creates many implications for responsibility, where an individual's evaluation of causing negative value to bring about positive value makes them responsible for both things.

This is a key distinction because it accepts the negative value of certain actions, without needing to justify them. Under most teleological perspectives, a negative action would be canceled out by a greater positive action. Under this view, removal of telephone operators would be canceled out by the billions of human-human interactions facilitated by automatic dialing, meaning there would be no responsibility implications for the negative value. When we look at non-teleological perspectives, our understanding of responsibility becomes clear and is not removed or reduced by outcome; rather, one must accept the consequences of

bringing about negative value, with the later positive value brought about being completely separate and not impacting the earlier negative value.

One may note that I have made an intentional choice not to argue for any particular value system, which is outside the scope of this thesis. This said, the correct value system must be able to navigate the challenges that the teleological view of value encounters, allowing for the intrinsic value of actions in their own right. This is extremely important because the teleological perspective can often be the default view of value, where we look toward pure efficiency as the sole metric for value. This is what directly results in mass layoffs and the replacement of human roles with automated ones. Once again, I am not stating that absolutely no values are teleological in structure, but rather, I argue for the prioritization of these non-teleological values in the decision-making process that augments human lives. This is so important because if careful consideration isn't given to the values of this non-teleological form, everything will be reduced to value based on outcome.

As computer scientists, we *must* intentionally consider Moore's understanding of value when we are thinking about what is valuable, and when we are comparing the values that will inform long-term decisions. My main thrust is as follows: even if computers can perfectly achieve all tasks we set them out to do, human-human interaction should never be valued any less; rather, it may move to different domains. We might not have a human ticket salesperson, but we would still have a human therapist. This means that all computational research *must* begin by examining the value of interactions through this lens, in order to guarantee the persistence of human-human interaction.

1.4 The Inherent Value of Cultural Diversity

The value that arises from cultural diversity is rooted in the benefits that result from including a wide range of perspectives. Essentially, there is something fundamentally wrong that occurs when we do not include a wide range of perspectives, or if we incorrectly characterize/unify these perspectives in a way that doesn't represent the values of the individual. This form of diversity brings to light the idea that there is *something* inherently valuable about the views, beliefs, customs, arts, etc. of people from different backgrounds. I define *culture* as the sharing of such values.

Within data for machine learning systems, cultural diversity can be valuable in two major ways: (1) in the data collection process, and (2) in the evaluation process. To better understand how diversity affects the machine learning process in these ways, I will distill some of the major findings that affected methodological approaches within the clinical sciences and apply these findings to computer science. I use the medical field to draw these connections to computer science, not because I think the former is prior to, or more important than, the latter. Rather, I think the conditions under which we conduct science in the medical field clearly reveal some of the deeper methodological implications for all scientific research. This is particularly true when it comes to the value of cultural diversity.

1.4.1 Methodological Issues in Clinical Science

It has been widely accepted that aspirin therapy reduces the risk of myocardial infarction (or heart attack) on average by 25 [40]; however, many of these early studies were heterogeneous in the composition of their participants. “Most of the participants in the 5 randomized trials were middle-aged men. Limited data are available to examine whether the effect of aspirin differs in other demographic groups, including elderly persons, women, and persons with diabetes or hypertension... Taken together, the trials involved more than 55 000 participants. Participants had a mean age of 59 years, and 86% were men.” [15].

This context makes clear why researchers did not fully understand the impact of demographics on the efficacy of aspirin until recently. For example, increasing evidence has shown that women have an increased likelihood of having aspirin resistance, making it less effective in women when compared to men [14, . When we compare studies that included women as part of their strata for stratified random sampling, we see a stark difference in efficacy. “The present study found that the gender mix of trials accounted for a substantial proportion of the variability in the reported efficacy of aspirin in reducing MI rates. Trials that recruited predominantly men demonstrated the largest benefits, while trials that recruited mostly women failed to demonstrate any benefits. These data are consistent with the notion that aspirin is less efficacious in women (for the reduction of MIs) and raise the possibility that women are more susceptible to aspirin resistance” [40].

Similar methodological flaws have occurred in numerous areas within the medical field, such as the realization that pulse oximeters are not able to properly function on patients with darker skin, or rather, that they must first be calibrated to function well. This flaw is fundamental to the physics of light transport, where higher amounts of skin pigmentation absorb more light rather than reflecting it, resulting in a less accurate measurement of blood oxygen using infrared light. This case is different because the flaw is grounded in physical limitations, rather than study design limitations; however, the same negative effects occurred. During the COVID-19 pandemic, individuals across the world were frequently using pulse oximeters as a means of health monitoring. As a result, accuracy concerns have been brought up regarding accuracy when used in people with different skin pigmentation. “The majority of the evidence suggests that the accuracy of pulse oximetry differs in subjects of different skin pigmentations to a level that requires particular attention, with decreased accuracy in patients with dark skin” [2].

I am not arguing against the physical limitations presented by a specific solution or technology. The rate of MI prevention in men versus women is determined by inherent genetic differences, and similarly, the efficacy of pulse oximeters on varying skin tones is largely determined by the physics of light transport. Rather, I am arguing for diversity as a way to understand a fuller picture of the effects of systems on individuals. The development of aspirin and the flaws underpinning pulse oximeters are simply model examples for ways in which a lack of diversity can impact the generalizability of research. Current research, especially in clinical science, places a strong focus on statistically valid sampling and external validation as a means of avoiding these issues.

The sampling method chosen to achieve statistical validity may vary based on the desired target application. With this in mind, findings that are aimed at generalization will often see researchers employing *stratified random sampling*. Stratified random sampling involves separating a population into smaller homogeneous, mutually exclusive groups called strata, from which the random samples are taken. This approach is used to ensure that, in expectation, the distribution of samples is balanced across strata. In other words, if the groups were separated by men and women, half of the samples would be from men and half of the random samples would be from women. This gives us the statistical guarantee that

resulting evaluations on the sample will have lower standard deviation and lower estimation error, allowing for results that are more likely to generalize to the full population.

In a similar manner, external validation involves applying the findings from a study to a different context. External validity is how we understand whether or not findings are generalizable, where generalizable findings will result in the same conclusions when applied to a broader population. It is important to note that external validation is not optional to make claims regarding generalizability: “External validation is necessary to determine a prediction model’s reproducibility and generalizability to new and different patients” [28]. In addition, results are typically encouraged to be validated by independent research, and to be validated multiple times and in different contexts to ensure the statistical significance of the results.

1.4.2 Epistemic Value

This concept of “understanding a fuller picture” is what I will call the *epistemic value* of diversity. Epistemology is broadly concerned with knowledge and how humans can know things. I will use the following definition of epistemology throughout this thesis:

Whenever a knower (S) knows some fact (p), several conditions must obtain. A proposition that S doesn’t even believe cannot be, or express, a fact that S knows. Therefore, knowledge requires belief. False propositions cannot be, or express, facts, and so cannot be known. Therefore, knowledge requires truth. Finally, S’s being correct in believing that p might merely be a matter of luck. For example, if Hal believes he has a fatal illness, not because he was told so by his doctor, but solely because as a hypochondriac he can’t help believing it, and it turns out that in fact he has a fatal illness, Hal’s being right about this is merely accidental: a matter of luck (bad luck, in this case). Therefore, knowledge requires a third element, one that excludes the aforementioned luck, and so that involves S’s belief being, in some sense, justifiably or appropriately held. If we take these three conditions on knowledge to be not merely necessary but also sufficient, then: S knows that p if and only if p is true and S justifiably believes that p. According to this account, the three conditions—truth, belief, and justification—are individually necessary and jointly

sufficient for knowledge of facts [33].

In summation, the *epistemic value* refers to how knowledge in the strictest sense - comprising truth, belief, and justification - can result in the increase of some evaluative aspect. This represents the concept of “understanding a fuller picture,” because the “picture” in question evaluates to truth. To connect things to the aspirin example detailed previously, testing the effects of aspirin on a largely male population *still* resulted in true information: the medication reduced the likelihood of heart attacks in patients. The key issue was that *despite* this true information, the studies did not represent the full truth, or rather, key aspects to the findings - such as the effect of demographics on prevention rates - were left out of the study. By contrast, exploring the effect of aspirin on women, the field as a whole gained a truer understanding of aspirin’s effect on *people* as a whole.

1.4.3 *The Epistemic Value of Cultural Diversity for Machine Learning*

To bring things together, the epistemic value of cultural diversity comes from the “fuller picture of the world” that is understood as a result of intentionally sampling from diverse cultures. Each of the methods used within clinical science to ensure the validity and generalizability of findings also applies to the machine learning process.

For machine learning systems using supervised learning, stratified sampling closely relates to the data collection process. In the same way that stratified sampling focuses on separating a population into homogenous, mutually exclusive groups (strata) to ensure balance before randomization, similar considerations are given to the process of collecting data and splitting it up into train, validation, and test sets. The goal behind every data split is to ensure that the validation/test set models the same distribution as the train set. If this is not the case, the results produced from the model after training will not model the *true* distribution of data. Once again, we see this theme of truth pushing its way to the foreground. Just like anything, the benefits of this strategy are highly context-dependent, but the majority of real-world machine learning applications need high levels of sample diversity to generalize. This becomes magnified when we try to model, or even probe a model’s understanding of, human values. By including culturally diverse data, we create a

truer representation of the values of *people*.

In the same way, external validation closely relates to the process of cross-validation in machine learning. Cross-validation occurs when we evaluate our model on any split that is not the training data. Once again, this validation/text data needs to be from the same distribution as our training data; however, if our distribution space is large, this could include a wide range of samples. This vast distribution space is typically where the most error is introduced when the cross-validation does not accurately represent the true distribution. An example of this is the FairFace dataset. Due to the sheer amount of data from white Western countries, the majority of public face datasets at the time were strongly biased toward Caucasian faces, while other races (e.g., Latino) were significantly underrepresented [21]. To fix this issue, Karkkainen and Joo developed a face dataset that was balanced across 7 races, and as a result, their dataset was able to achieve much better generalization classification performance for gender, race, and age on novel image datasets collected from Twitter, international online newspapers, and web search, which contained more non-White faces than typical face datasets. In addition, the model trained on their dataset produced balanced results across race [21]. This dataset employed the benefits of stratified sampling, while also becoming an external validation source for different models. Since their dataset was balanced across the aforementioned demographic factors, it was a more faithful representation of the true underlying data distribution, allowing for better generalization on unseen data.

The broader lesson is that diversity is not merely a social or political consideration; it is an epistemic one. A dataset that systematically excludes particular populations, cultures, or value systems necessarily provides an incomplete picture of the world. Any model trained on such data will inherit these limitations, producing representations that reflect only a subset of human experience. Conversely, increasing cultural diversity within a dataset expands the range of phenomena available for learning and evaluation, improving both the validity of empirical conclusions and the robustness of model behavior. In this sense, cultural diversity functions as a mechanism for reducing epistemic blind spots. It enables machine learning systems to construct models that more closely approximate the complexity of the populations they are intended to serve.

These considerations become especially important when the target of modeling is not physical phenomena but human values themselves. Unlike many perceptual tasks, values are deeply embedded within social, historical, and cultural contexts. A dataset that captures only a narrow segment of humanity’s value judgments risks conflating local norms with universal truths. Consequently, cultural diversity is not simply beneficial for developing value-sensitive machine learning systems; it is a prerequisite for any credible attempt to model, evaluate, or reason about human values at scale.

Chapter 2

IN DEFENSE OF MORAL REALISM: THE POWER OF RATIONAL REFLECTION

2.1 Introduction

The metaethical question of moral realism is the foundation for how discussions on morality ought to be conducted. This is because aligning with the theory of moral realism or antirealism embeds many implicit notions within any subsequent discussion of morality. In short, moral realism asserts that morality is a fact and corresponds to some truth in the world. Things may or may not align with “the right,” but what separates the realist from the antirealist is that the realist holds to the idea that “the right” exists. Note that this holds no assumptions regarding whether or not one can ascertain “the right” or the source of “the right.” Those statements would fall under different subcategories, but the core difference comes from their views on the real existence of morality.

Within this thesis, I assert moral realism; however, the goal of this work is not to provide a comprehensive argument for realism. Rather, this chapter focuses on exposing a major flaw in a set of antirealist theories called Evolutionary Debunking Arguments (EDAs). By exposing this flaw in EDAs (at least those of the form of Sharon Street’s *Darwinian Dilemma*), I aim to provide a defense for moral realism without diving fully into the literature. This chapter will suffice to give the reader a reasonable overview of the discussion of moral realism, and will serve to ground one’s understanding of my views as I continue to discuss the moral implications of computing.

Within the debate surrounding Evolutionary Debunking Arguments (EDAs), the argument proposed by Sharon Street within *A Darwinian Dilemma for Realist Theories of Value* emerges as a major work aiming to debunk the theory of moral realism through an evolutionary approach [34]. Within such arguments, *reflective equilibrium* is a commonly used term typically used to describe the state of a person’s beliefs after rational reflection.

A major issue, however, is that this term is often invoked without a consistent definition.

John Rawls, the philosopher who first coined the term, separated reflective equilibrium into two forms, *wide* and *narrow* reflective equilibrium. Each has distinct limitations that complicate their use in philosophical arguments, particularly in Street’s EDA. I argue that Street’s characterization of reflective equilibrium is invalid: she claims that our initial evaluative judgments constrain the outcome of reflective equilibrium, thereby limiting the scope of rational moral reflection.

This work argues that such a constraint is unjustified. A truly rational reflective agent must be able to arrive at evaluative judgments that are not merely shaped or contaminated by initial intuitions, tendencies, or proto-beliefs. To support this claim, I present a method by which a rational agent, operating within Street’s own framework, can transcend the limits she proposes and arrive at judgments that are genuinely independent of initial biases.

2.1.1 *Street’s Reflective Equilibrium*

The core of Street’s argument focuses on debunking the realist theory of value. The realist view of value is that “there are at least some evaluative facts or truths that hold independently of all our evaluative attitudes” [34, p. 110], where evaluative attitudes are later defined to include desires, attitudes of approval and disapproval, unreflective evaluative tendencies, and conscious/unconsciously held evaluative judgments. This definition is central to her argument because the EDA explores how our evaluative attitudes influence our evaluative judgments, especially with regard to the set of possible judgments upon rational reflection. Specifically, “Where anti-realists part ways with realists is in denying that there are evaluative truths which hold independently of the whole set of evaluative judgments we make or might make upon reflection, or independently of the whole set of other evaluative attitudes we hold or might hold upon reflection” [34, p. 110].

The anti-realist view focuses on reflection as a means by which an agent can explore the space of possible evaluative judgments, moving past their initial unreflective evaluative tendencies. My argument does not aim to further explore or deny the anti-realist position; rather, I aim to examine and modify how Street has defined reflective equilibrium to

understand how this augments her argument.

Street's definition of reflective equilibrium is clearly presented within the following passage:

“The widespread consensus that the method of reflective equilibrium, broadly understood, is our sole means of proceeding in ethics is an acknowledgment of this fact: ultimately, we can test our evaluative judgements only by testing their consistency with our other evaluative judgements, combined of course with judgements about the (non-evaluative) facts. Thus, if the fund of evaluative judgements with which human reflection began was thoroughly contaminated with illegitimate influence - and the objector has offered no reason to doubt this part of the argument - then the tools of rational reflection were equally contaminated, for the latter are always just a subset of the former. It follows that all our reflection over the ages has really just been a process of assessing evaluative judgements that are mostly off the mark in terms of others that are mostly off the mark... So long as we assume that there is no relation between evolutionary influences and evaluative truth, the appeal to rational reflection offers no escape from the conclusion that, in the absence of an incredible coincidence, most of our evaluative judgements are likely to be false” [34, p. 124-125].

The goal behind this line of reasoning is that she aims to show how our unreflective evaluative tendencies influence all our evaluative attitudes, even those within reflective equilibrium. Succinctly put, “My point here is instead the simple and plausible one that had the general content of our basic evaluative tendencies been very different, then the general content of our full-fledged evaluative judgements would also have been very different...” [34, p. 120]. This builds to the core of her EDA, the two-horned 'Darwinian dilemma' for value realism.

Street writes that if our evaluative attitudes were to track evaluative truth, then one of two things must have happened:

1. It occurred by pure coincidence.
2. By some “tracking account,” evolutionary forces allowed our evaluative attitudes to track evaluative truth.

I focus on the second. The objection she brings to this horn asserts that by the “adaptive link account,” our evaluative tendencies must be entirely explained by evolutionary factors. This account is combined with her definition of reflective equilibrium to form her refutation of the horn. She describes that the space of attainable evaluative attitudes for a rational reflective agent is limited by the adaptive link account.

It is important to note that Street does understand that our rational capabilities allow us to step back from a problem and consider it from different angles. “...I do not mean that we automatically or inevitably accept the full-fledged evaluative judgements that line up in content with our basic evaluative tendencies. Certainly not. For one thing, other causal influences can shape our evaluative judgements in ways that make them stray, perhaps quite far, from alignment with our more basic evaluative tendencies. For another thing, we are reflective creatures, and as such are capable of noticing any given evaluative tendency in ourselves, stepping back from it, and deciding on reflection to disavow it and fight against it rather than to endorse the content suggested by it” [34, p. 120]. A major issue is that Street’s definition of reflective equilibrium is not consistent with John Rawls’ original definition of the term, as I will continue to explore.

2.2 Rawls’ Reflective Equilibrium

John Rawls first coined the term “reflective equilibrium” in the paper *Justice as Fairness*. He describes it as the following: “By going back and forth, sometimes altering the conditions of the contractual circumstances, at others withdrawing our judgments and conforming them to principle, I assume that eventually we shall find a description of the initial situation that both expresses reasonable conditions and yields principles which match our considered judgments duly pruned and adjusted. This state of affairs I refer to as reflective equilibrium” [30, p. 20]. By this definition, Rawls specifically clarifies that this form of reflective equilibrium simply aims to achieve internal coherence among our principles and judgments; however, he adds that it may not necessarily be stable.

This definition of reflective equilibrium thus far does not conflict with Street’s main line of argument, since she primarily aims to show how our unreflective evaluative tendencies influence our later evaluative judgements. The contention comes when Rawls expounds upon

this definition. He explores the implications embedded within the instability of reflective equilibrium. For a rational reflective agent, the equilibrium is liable to be upset by further examination or additional information, which could lead to a revision of judgment. With this instability, the core idea behind rational reflection is to justify our own convictions.

“When a person is presented with an intuitively appealing account of his sense of justice (one, say, which embodies various reasonable and natural presumptions), he may well revise his judgments to conform to its principles even though the theory does not fit his existing judgments exactly. He is especially likely to do this if he can find an explanation for the deviations which undermines his confidence in his original judgments and if the conception presented yields a judgment which he finds he can now accept” [30, p. 48].

Rawls states that through rational reflection, a moral agent may augment their unreflective evaluative tendencies to favor another judgment. Here, he describes an intuitively appealing judgment, but it doesn’t necessarily need to be so. He puts it more clearly in his later work, *The Independence of Moral Theory*: “Although in order to get started various judgments are viewed as firm enough to be taken provisionally as fixed points, there are no judgments on any level of generality that are in principle immune to revision” [30, p. 8]. Under the Rawlsian reflective equilibrium, every judgement may be revised, even those that start as being influenced by unreflective evaluative tendencies. This is an extremely important distinction because Street’s reflective equilibrium bakes in the assumption that the space of possible evaluative attitudes for a rational reflection agent cannot be synonymous with the space of all possible evaluative attitudes due to the limits of the agent’s initial unreflective evaluative tendencies.

Despite these characteristics, Rawls goes on to acknowledge that there are multiple interpretations of reflective equilibrium. “For the notion varies depending upon whether one is to be presented with only those descriptions which more or less match one’s existing judgments except for minor discrepancies, or whether one is to be presented with all possible descriptions to which one might plausibly conform one’s judgments together with all relevant philosophical arguments for them” [30, p. 49]. These two notions are what he later separates into *wide* and *narrow reflective equilibrium*. Narrow reflective equilibrium is when one just aims to reach internal coherence among their beliefs, while wide reflective equilibrium is

where the agent brings in many other considerations (specifically those that are very different from one’s existing judgments), ideally, all possible considerations.

Street’s definition was much closer to narrow reflective equilibrium; however, Rawls expressly did not view narrow reflective equilibrium as enough to do more than smooth out bumps in the internal coherence of a belief. Despite this disagreement, even Rawls was skeptical regarding the practicality of wide reflective equilibrium: “To be sure, it is doubtful whether one can ever reach this state. For even if the idea of all possible descriptions and of all philosophically relevant arguments is well-defined (which is questionable), we cannot examine each of them” [30, p. 49]. In this statement, Rawls agrees with Street, where “The most we can do is to study the conceptions of justice known to us through the tradition of moral philosophy and any further ones that occur to us, and then to consider these” [30, p. 49]. Essentially, we are limited by that which we have/can comprehend. I argue this is not necessarily true, and for certain subclasses of problems, a rational reflective agent can feasibly consider all possible conceptions within the space of potential evaluative attitudes, resulting in an evaluative judgement that is not impacted by initial unreflective evaluative tendencies. I will, in turn, show how this impacts Street’s EDA.

2.3 *The Limits of Rational Reflection*

To understand how I refute Street’s formulation of reflective equilibrium, we represent her definition as a Markov process, where each decision made by agent A is an event within the continuous-time Markov chain. We can effectively utilize Markov chains for this formulation because each step within reflective equilibrium could be considered a stochastic process. Before the reflective process begins, a rational reflective agent must have some probabilistic likelihood of selecting evaluative judgements within the space of all possible judgements, even if the likelihood is zero. Markov chains focus on the likelihood of a future event occurring based on the current state, and this is exactly the process described by both Rawls and Street. In addition, the Markov chain is continuous-time because the process of reflective equilibrium is hardly discrete, meaning that there are not always set start and end times to a process of rational reflection.

To effectively consider Street’s argument, I break it down into its formal logic:

- Let A be a rational reflective moral agent.
- Let $p \in \mathcal{P}$ be the current value assignment proposition sampled from the set of all possible value assignment propositions $\mathcal{P}$.
- Let $t \in [0, \infty)$ be the current time-step
- Let α_p be the space of all possible evaluative attitudes for the given proposition p . Note both α and p are considered to be within the same space, but α may be aligned or misaligned with p .
- Let $Y_t \in \alpha_p$ be the evaluative attitude for agent A held at time t , where Y_t must be within α_p .
- Let $X_t \in Y_t$ be the evaluative judgment made at time t , where X_t must be within agent A 's evaluative attitudes at time t .
- Let $(X_t)_{t \geq 0}$ be a continuous-time Markov chain of evaluative judgments made *within the process of rational reflection* over proposition p .

Note for these definitions, I reference Street's original explanation of the terms: "Evaluative attitudes I understand to include states such as desires, attitudes of approval and disapproval, unreflective evaluative tendencies such as the tendency to experience X as counting in favor of or demanding Y , and consciously or unconsciously held evaluative judgements, such as judgements about what is a reason for what, about what one should or ought to do, about what is good, valuable, or worthwhile, about what is morally right or wrong, and so on" [34, p. 110]. It is important to make the distinction that α is simply the space of all possible evaluative attitudes for the problem, not for agent A . This means that there may be evaluative attitudes contained within α_p that are impossible for A to consider.

Given the above, Street makes the following assertion:

$$\forall p \forall A \forall t \geq 0, X_t \perp X_0 \implies \mathbb{E}[X_t] = 0$$

In other words, Street states that any evaluations in contradiction to the initial unreflective evaluative tendency X_0 cannot be selected due to the impact evolutionary forces have on our tendencies. For this, I once again reference Street's writing, "Thus, if the fund of evaluative judgements with which human reflection began was thoroughly contaminated with illegitimate influence... then the tools of rational reflection were equally contaminated..." [34, p. 124].

For the sake of argument, I focus on the second statement, "then the tools of rational reflection were equally contaminated." With Street's formulation thoroughly described, I begin my refutation by clarifying what it means for there to exist a rational reflective agent A . For a value-assignment proposition p that has not been considered (so we start at time $t = 0$):

1. A rational reflective agent A must be equally likely to select given evaluative judgments X or Y if A evaluates them to have equal value. For example, a student would be equally likely to judge test X or test Y as easy if they were confident in all the material from the course.
2. A fair rational reflective moral agent A must have some non-zero likelihood of selecting evaluative judgement X if it has not been evaluated, or if given an adequate reason. Furthermore, A must be more likely to select judgment X rather than Y if they deem the rationale for X to be stronger than that for Y . This clause simply clarifies that a fair agent should be open to making an informed decision before starting the rational reflection process.
3. A rational reflective agent A must have a zero likelihood of selecting an already considered and rejected evaluative judgement X without the presence of new evaluative information.
4. A rational reflective agent A must assign equal value to evaluative judgements that are not relevant to A

To refute Street’s original statement, I only need to show that there exists an agent A that can rationally come to a judgment that is in direct contradiction to their original evaluative tendency. In other words:

$$\exists p \wedge \exists A, \exists t \geq 0 \text{ such that } (X_t \perp X_0) \wedge (\mathbb{E}[X_t] \neq 0)$$

My argument roughly follows this structure:

- Premise 1: A rational reflective agent can make an evaluative judgment that directly contradicts their initial unreflective evaluative tendencies.
- Premise 2: A rational reflective agent can explore the full space of evaluative attitudes α (i.e., wide reflective equilibrium).
- Conclusion: Therefore, a rational reflective agent can arrive at an evaluative judgment in wide reflective equilibrium that contradicts their initial unreflective evaluative tendencies.

Considering a value assignment proposition p , let there exist some unbiased rational reflective moral agent A .

To address premise 1, I focus on clause 2 of the definition of rational reflective moral agent. Specifically, if A is provided a reason that they deem to be strong enough, then they can be convinced that a different judgment is correct. For example, let’s say the proposition p is “eating 100 pounds of ice cream is good for you.” We are given that agent A is a child who, due to evolutionary pressures, ended up with taste buds that make it more likely for them to evaluate foods with high sugar content as tasting good. The child’s initial unreflective evaluative tendency is that sweet things taste good, and whatever tastes good must be good for them; however, in the future, they would likely learn about the health issues that could be caused by eating so much ice cream so quickly. Thus, they would change their evaluative judgment without changing their tendency to think sweet things taste good.

Street is actually fully on board with this premise. “ultimately, we can test our evaluative judgements only by testing their consistency with our other evaluative judgements,

combined of course with judgements about the (non-evaluative) facts” [34, p. 124]. Non-evaluative facts are valid ways of examining our judgments, and thus, are one way to form a judgment that contradicts initial unreflective evaluative tendencies. The disagreement comes in premise 2.

To address premise 2, I focus on the space of all potential value assignment propositions $\mathcal{P}$. For any value assignment proposition p , there is an implicit relationship, where p only evaluates to true if $\forall q \in \mathcal{P}$ such that $q \neq p, p \wedge \neg q$. This is what I call the independence of proposition space. This distinction is important because within proposition space, there exist no subsets; rather, there are similarities. One can think of it like a mathematical unit vector space, where each proposition is a vector that could be compared to another vector by means of cosine similarity. This also means that a proposition like “kicking kittens is always wrong” is not a superset of the proposition “kicking kittens is sometimes wrong.” They are independent propositions that are similar in their value assignment.

We can use another example to further understand the independence of proposition space. Given the proposition, “it is always good not to murder your children”, p corresponds to endorsing the proposition that refraining from such an act is morally good. This implies the rejection of all alternative moral framings q that contradict, deny, or are indifferent toward that position because they are logically incompatible with p , and thus are excluded by $p \wedge \neg q$. An invalid q would be something like “it is morally neutral to murder your children.” By the logical definition above, “it is always good not to murder your children” and “it is not morally neutral to murder your children” are logically consistent and so the statement “it is morally neutral to murder your children” would be invalid. One can think of this as a “rather than” relationship, where we are stating X is permissible rather than Y. Note that the independence of proposition space does not make any claims as to the truth or falsity of p , it solely maps p in terms of all other propositions within $\mathcal{P}$.

Finally, for premise 2, I must clarify what is meant by “evaluation” through “rational reflection”. For rational reflection to occur, A must consider the reasons why p . One could argue that A must also consider all the reasons why p rather than q ; however, due to the independence of p as defined above, such a consideration is not necessary. Therefore, evaluative judgement X_t is a valid evaluative judgement through rational reflection, if and

only if it is formed by a proposition p and justified by some rationale r . Note that this rationale does not need to be strong, but *reason* must *by definition* be used within rational reflection. With this in mind, I update my set of definitions:

- Let $\mathcal{R}$ be the set of all **relevant** rationales for proposition p . Similarly $\mathcal{R}_n$ be the $\mathcal{R}$ at pass n when updating p .
- Let $r \in \mathcal{R}$ be a rationale from $\mathcal{R}$
- Let $X_t \in Y_t$ such that $X_t = r \vdash p$ be the evaluative judgment made at time t , where X_t must be within agent A 's evaluative attitudes at time t , and such that X_t is the true evaluation of proposition p as justified by rationale r .

With this definition of “evaluative judgment by means of rational reflection”, we start to see the challenges Rawls faced when examining wide reflective equilibrium. “To be sure, it is doubtful whether one can ever reach this state. For even if the idea of all possible descriptions and of all philosophically relevant arguments is well-defined (which is questionable), we cannot examine each of them” [30, 49]. The full space $\mathcal{R}$ is intractable for a realistic agent A to evaluate, since it is theoretically countably infinite. Despite this, one aspect of the space that was unexplored by both Rawls and Street is the notion of relevance.

A rationale r is “relevant” to agent A **if and only if** X_t would be affected (either increased agreement or disagreement) as a result. This means that not all rationales are relevant to A . For instance, given the proposition “kicking kittens is wrong,” the rationale “because I said so” would *likely* have a lower likelihood of being selected than “because it introduces unnecessary suffering into the world.” This example creates numerous implications regarding how to quantify the likelihood of a rationale being selected; however, this is irrelevant to my argument. Rather, I solely aim to show that certain rationales have a zero likelihood of being selected because they are irrelevant.

Since the space of $\mathcal{R}$ is countably infinite, according to clause 4 of the definition of rational reflective agent, the likelihood of selecting each rational will approach zero. Mathematically:

$$\lim_{i \rightarrow \infty} \mathbb{E}[r_i] = 0$$

Therefore, if we are given a limited proposition p such that the majority of possible decisions are irrelevant, the space $\mathcal{R}$ is countably finite and agent A can feasibly engage in wide reflective equilibrium. In order to form this proposition, one simply needs to embed a rationale within p such that $p_2 = r \vdash p_1$. This works because the propositions and evaluative judgments are from the same space; a judgment simply affirms or denies a proposition. For example, given the proposition “kicking kittens is bad,” the space of all possible rationales $\mathcal{R}_p$ is intractable, because the infinite set

$$\{\text{“because I said so”, “because it introduces unnecessary suffering”, ...}\}$$

are all valid rationales, no matter their strength. Despite this, I can redefine the proposition to include one of the rationales, thereby limiting the space of relevant evaluative judgments that can be made.

An example of the new p would be “kicking kittens is bad because it introduces unnecessary suffering.” We can effectively add this rationale to p because if A was already likely to agree with the statement “kicking kittens is bad,” then “because it introduces unnecessary suffering” would only increase their agreement with p so it is a relevant rationale. With this updated p the new space of relevant rationales is even more limited and might look like

$$\{\text{“and because suffering is bad”, “and because unnecessary suffering is gross”, ...}\}$$

Note that for the second pass of relevant rationales, we may not use any rationale that was considered for the first pass. This is due to the fact that, by definition of *relevant*, the rationale would not augment A ’s judgment toward the new proposition p .

These distinctions are important because recall that for my argument to work, I only need to show that there exists a proposition that satisfies the following criteria:

$$\exists p \wedge \exists A, \exists t \geq 0 \text{ such that } (X_t \perp X_0) \wedge (\mathbb{E}[X_t] \neq 0)$$

Therefore, I am able to redefine the proposition p to include the rationale r while still rejecting Street’s description of the limits of rational reflection. Thus, I have shown that there exists a rational reflective agent that can arrive at an evaluative judgment in wide reflective equilibrium, which contradicts their initial unreflective evaluative tendencies.

2.4 *An Answer to the Possibility of Influence*

A major objection to my line of argument ties back to Street's original formulation. The objection goes as follows: for Street's argument to get off the ground, she doesn't need to show that one's final state is *necessarily* influenced by our initial unreflective evaluative tendencies; she only needs to show that it *can* be influenced by these tendencies. The distinction is important because if she aimed to show that one's final state is necessarily influenced by initial tendencies, then she must show this for all possible propositions p in the proposition space $\mathcal{P}$.

The primary issue is that this objection produces the skeptic's conclusion to the Darwinian dilemma. If Street only aimed to show that one's final state *can* be influenced by initial tendencies, then the conclusion would be to plant a seed of doubt that one's judgments are completely unbiased from their initial tendencies. Despite this, Street's argument is far more than a skeptic's conclusion. As an EDA, the argument aims to debunk the realist theory of value by showing the necessity of influence. "My conclusion: the content of human evaluative judgements has been tremendously influenced - indirectly influenced, in the way I have indicated, but nevertheless tremendously influenced - by the forces of natural selection, such that our system of evaluative judgements is saturated with evolutionary influence. The truth of some account very roughly along these lines is all that is required for the Darwinian Dilemma to get off the ground" [34].

Therefore, my response to this objection is quite simple: Street's original goal was to show that one's final state is necessarily influenced by initial unreflective evaluative tendencies. Once again, "Thus, if the fund of evaluative judgements with which human reflection began was thoroughly contaminated with illegitimate influence... then the tools of rational reflection were equally contaminated" [34]. This clearly shows that Street's focus was to show that all humans have had their tools of rational reflection contaminated; therefore, their future judgments must be influenced by the initial tendencies. She does this because her goal is to show that if any evaluative judgement can be influenced by initial tendencies, then we may lose confidence in the independence of all our evaluative judgements from these tendencies. This is how she forms the Darwinian dilemma. If one's final state

can be influenced by initial judgments, then the only way for the realist to have confidence that their evaluative judgments are truth-tracking would be to have some tracking account; otherwise, natural selection would be considered purely a distorting force on our judgments.

2.5 *An Answer to the Independence of Future Judgments*

A second major objection to my argument relies on Street’s original characterization of the influence of unreflective evaluative tendencies. Essentially, this contests that even if one were able to consider all possibilities within the proposition space through wide reflective equilibrium, their initial unreflective evaluative tendencies would influence any judgments they come to. This “corruption,” as Street labels it, eliminates our ability to make a completely unbiased judgment.

This objection is important because it would mean that Street does not need to care whether the agent engages in wide or narrow reflective equilibrium. All of their future judgments would have been determined by initial tendencies. To take this line of reasoning to the extreme, one could conclude that any arguments about the process of reflection are meaningless, due to the overwhelming influence of initial tendencies.

To answer this objection, I turn back to how I defined a rational reflective agent, specifically clause 2 of the definition, “A fair rational reflective moral agent *A* must have some non-zero likelihood of selecting evaluative judgement *X* if it has not been evaluated, or if given an adequate reason.” The extra term “fair” is only added here because neither Street nor Rawls explicitly requires that a moral agent be fair; however, to counter Street’s EDA, I only need to show that an agent *can* escape the dilemma. If an agent is fair, then their rational reflection should fully consider relevant rationales in a manner that is completely removed from their initial evaluative tendencies.

If Street’s argument was “had the general content of our basic evaluative tendencies been very different, then the general content of our full-fledged evaluative judgements would also have been very different...” [34], then my counterargument is that for a fair rational reflective agent, the content of their basic evaluative tendencies do not ultimately decide their future judgments. Rational reflection has much more evaluative power than Street credits.

2.6 *What About the Darwinian Dilemma?*

Given the possibility of a rational reflective agent being able to enter wide reflective equilibrium, Street's Darwinian dilemma now must account for such a scenario. Evolution can no longer be considered to be a purely distorting force, or rather, it cannot be considered to be an irreparable distortion. The ability for a rational reflective agent to form evaluative judgments through which their initial unreflective evaluative tendencies have no effect shows that rational reflection is able to offset the influence of evolutionary pressures. I have not made any claims as to the truth-tracking ability of such reflection, which is outside the scope of this paper. Rather, for Street's argument to be strong, she must address how rational reflection can eliminate the impact of evolutionary pressures (as she describes them) for the class of propositions as described within this paper.

Chapter 3

MORAL INFLUENCE

3.1 Introduction

Images are a central aspect of this discussion on morality and value. Images are powerful because they can carry moral values. Perspectives may vary regarding this claim. How can images carry moral value if they are just ink on a page, or pixels on a screen? In this chapter, I will defend the claim that images can carry intrinsic moral value and that this moral value has a direct link to the moral reactions of moral agents.

3.2 The Intrinsic Moral Value of Media

In the prior chapter, I discussed what it means for something to be intrinsically valuable. To distill these distinctions, I leveraged G.E. Moore’s conception of value: “In order to arrive at a correct decision on the first part of this question [‘What things have intrinsic value, and in what degrees’], it is necessary to consider what things are such that, if they existed by themselves, in absolute isolation, we should yet judge their existence to be good” [27, p. 187]. This framing makes clear the fact that the *intrinsic* value of a thing cannot rely on external factors. In the case of actions, this meant that neither intent nor the outcome of an action could influence that which had intrinsic value (so the act of “punishing the innocent” intrinsically had negative value).

To understand how this framing of value affects how we view the intrinsic value of media, let’s start by exploring the intrinsic value of images. The same learnings from considering the intrinsic value of actions (and by nature interactions as well) can be applied to our understanding of the intrinsic value of images. If I were to ask the question, “Can images have intrinsic value?” most would reply yes; however, the response becomes more nuanced when I ask the question, “Can images have intrinsic value?” The nature of something being intrinsic affects things because it means that the evaluation can no longer rely on external

factors for influence. I argue that images *can* have intrinsic value, and it is directly related to the earlier discussion of whether human-computer interaction can have intrinsic value.

Within that discussion, we determined that human-computer interaction can have intrinsic value, but the nuance arose when we examined two different forms of interaction with computers. The first was interaction *through* (or mediated by) a computer, while the second was interaction *with* a computer. These two forms of interaction were different within our discussion of value, because the focus of the interaction augmented the resulting evaluation. Even though interaction *through* a computer required interaction *with* a computer, the two were evaluated separately. For example, the value of texting friends focuses on the connection and interaction between friends rather than the technology of texting. We could examine the value of text messaging and come to a result that is distinct from the associated value of texting friends.

This directly relates to the intrinsic value of images because, similarly, we can refer to images that either *carry* value or images that can *be* valuable. To clarify, when I discuss images in the context of this thesis, I am specifically referencing the technology of creating a two-dimensional representation of something that we can perceive through vision. Images do not need to reproduce something that exists in the real world, but I am focusing on images as strictly visual media. With this in mind, if an image can *be* valuable, that means that the fundamental technology of creating the visual representation has value in and of itself. While some might argue for or against this point, the distinction makes clear the fact that it is unrelated to our discussion on “moral images.” The idea of a moral image embeds the notion of one needing to first view the image to evaluate its rightness or wrongness. This need to view the image in order to assess its value exposes what we mean when we’re asking if an image can have intrinsic value. We are actually asking, “Can images carry intrinsic value?”

To this question, I argue yes, images can carry intrinsic value. Once again, my justification is for the same reasons why computers can mediate human-human interaction. When we ask if an image can carry intrinsic value, we are asking whether an image can *represent* something that has intrinsic value. For this to be true, we need only examine whether certain things have intrinsic value and whether images can adequately represent them. In

the previous chapter, I established that certain actions can have intrinsic value, so now we need to think about whether an image can create a “good enough” representation of those actions that have intrinsic value.

The quality of an image’s representation of actions directly corresponds to the state associated with that action. Frequently, this state can carry important contextual information that is difficult to represent visually. For example, it might be difficult to *visually* represent whether a person is innocent or guilty within the scenario “punishing an innocent person.” This said, this doesn’t mean that it is impossible to visually represent actions that have intrinsic value; it just means that the set of actions with intrinsic value that can be *purely* visually represented is a subset of the set of actions that have intrinsic value. When examining the scenario “kicking babies,” one could reasonably assert that this scenario had intrinsic negative value, no matter the modality of its representation. This properly showcases my final justification for why images can carry intrinsic value. Images carry intrinsic value when the situation or action depicted within the image would be evaluated to be positive or negative if examined in isolation, **regardless of modality**.

The characteristic of intrinsic value within images being modality agnostic is crucial to how G. E. Moore formulates his method of assessing value. The modality-agnostic nature of this definition also means that the approach to asserting that images represent intrinsic value can be applied to media of any modality. For instance, text can carry intrinsic value when it adequately represents something that has intrinsic value. Audio can carry intrinsic value when it adequately represents something that has intrinsic value. Even physical media can carry intrinsic value. This brings to light the core definition of “media,” as being that which mediates or brings about something else (often used in the context of communication). Media *fundamentally* is defined as carrying out/mediating something else. Therefore, in addition to *being* a communication of a representation of reality, media can also *be* a communication of value. Since these aspects are core to the being of media, it means that media can have intrinsic value.

By establishing the intrinsic value of, it becomes clear why I argue media can also be moral. Moral values are specifically values that are related to “the right” or “the wrong.” These are often related to “the good” and “the bad,” but they are more specific. This

is often clarified when put in the context of normative moral theory. Normative moral theories aim to describe/characterize what people *should* do. This concept of *should* carries a significant amount of weight, especially when viewed in the context of actions. Under a normative moral theory, certain actions and interactions are not only good, but they are *right*. In other words, everyone ought to strive for such occurrences.

3.3 From Moral Influence to Moral Reactions

3.3.1 How Moral Media Brings About Moral Reactions

With this understanding that media can carry *intrinsic* moral value, we must explore the connection between the moral value of media and how this value can influence the values of individuals. Within this discussion of moral reactions, I would like to preface by stating that moral reactions need not be solely brought about by media; however, media is an effective and portable method to demonstrate the connection in the ways that intrinsic moral value can influence the values held by individuals.

The most powerful proof of this connection comes through perception. Please view the following image and take note of what reactions arise.

(**content warning:** graphic wartime imagery; visible burn injuries and child suffering).



Figure 3.1: The Terror of War (also known as “Napalm Girl”), June 8, 1972: Kim Phúc, centre, running down a road after a South Vietnam Air Force napalm attack [36]

What reactions arose upon viewing this photo? Shock, sadness, and/or anger are all likely natural reactions to the atrocities depicted in this photo, which over time has come to be known by the title “Napalm Girl.” This photo was taken in 1972 and went on to receive the Pulitzer Prize and World Press Photo awards. These awards were due to the impact that was had on the public perception and opposition to the Vietnam War. This is a powerful example of the way that visual imagery can be a powerful tool. When we dissect these reactions even further, it becomes clear that the image **caused** the reaction. The reaction may have been latent, or it may have been a more prominent expression of your internal values, but this causal relation between the image and the resulting reaction is a core aspect of how we interact with morally salient media daily.

The ability of imagery to not only carry intrinsic moral value but also to cause moral reactions leads to its frequent use within fields such as human rights advocacy, where the negative emotions can cause larger systemic change. In ways, these negative emotions are leveraged and utilized to bring about stronger motivation than pure description ever could. The power of moral reactions is where phrases such as “seeing is believing” or “a picture is worth a thousand words” originate, because initial reactions are powerful. To utilize the language of previous chapters, these initial moral reactions correspond to one’s pre-reflective evaluative tendencies.

Our understanding of pre-reflective evaluative tendencies is built upon this causal relation. The fact that something intrinsically valuable can *cause* a moral reaction means that we can understand our pre-reflective evaluative tendencies through this lens of moral reactions to moral media. That which produces positive and negative moral reactions can cut to the heart of our *initial* connection to right and wrong. Despite this interplay, it is still unclear how our moral reactions correspond to potentially contrasting moral outcomes. In a similar vein, I have argued that moral value is not related to outcome; however, there are often cases where moral outcome can influence how we think about moral value.

3.3.2 *How Moral Reactions Define Moral Influence*

The “Napalm Girl” photo, which depicts a scene that is aligned with “the bad” and thus carries negative moral value, was able to produce a morally positive outcome. This contrast in values is why individuals are often hesitant to label the photograph as morally positive or negative, because there is a conflation in the semantics that the term *moral value* can take on. With this in mind, my justification in the previous section defines moral value (and specifically intrinsic moral value) as that which is carried by the thing *itself*. This showcases the distinction between moral value and moral influence. **Moral influence** corresponds to the moral outcomes that are caused by a thing. In this way, the distinction becomes clear. *Moral value is not teleological in form, while moral influence is.*

This is how moral influence directly relates to moral reactions. Moral reactions can impact future actions, which impact future outcomes, which is how moral influence is defined. Thus, the moral reactions caused by media can lead to the moral influence of that media. It is important not to conflate the colloquial understanding of wrongness with intrinsic wrongness. To specifically utilize the example of media, media that is intrinsically wrong can produce right outcomes, but this does not mean that that which is being mediated is right. Therefore, the “Napalm Girl” photo is intrinsically morally wrong because when examined in isolation, it mediates that which would be evaluated as wrong. By contrast, the moral influence of the photo would be morally right (or at least right in certain ways) due to the widespread positive impact. We have just examined how negative moral reactions can bring about positive moral influence, but the relation between moral reactions and moral influence need not always be inverse. There can be a direct positive or negative relation, in the same way that there can be an inverse positive or negative relation. This myriad of relations becomes clear when we examine how media can be used negatively.

Since moral media can cause moral reactions, this also means that it can be used to manipulate and deceive. For example, the same “Napalm Girl” photograph that helped generate public opposition to the Vietnam War could also be employed in a different context to produce harmful outcomes. An image’s moral influence is not fixed by its intrinsic moral value. Rather, it depends on the social, political, and psychological effects that the image

has on those who encounter it. The power of morally salient media, therefore, lies not only in what it depicts but also in its ability to shape the reactions of an audience. This is precisely why images have historically occupied a central role in political propaganda, advertising, and human rights advocacy. The emotional force of visual media can be leveraged to motivate moral action, but it can equally be leveraged to reinforce prejudice, hatred, or misinformation.

This duality becomes particularly evident in the context of human rights advocacy. As Hopgood notes, representations of suffering women and children frequently become symbolic embodiments of innocence and vulnerability, serving as powerful reminders of human wrongdoing. “But the suffering child also embodies the central figure of liberal humanism: The natural individual. Innocence is a proxy for naturalness (guilelessness, blamelessness). This ‘universal essence,’ or ‘species existence,’ identifies a human being as ‘without differentiation or distinction in his nakedness and simplicity, united with all others in an empty nature deprived of substantive characteristic.’ In this way both compassion and justice can be anchored on the child. Nothing is more authentic” [17, p. 71]. The suffering child, in particular, functions as a universal moral subject capable of eliciting compassion across cultural and political boundaries. Such imagery can motivate humanitarian action by creating an immediate emotional connection between observer and victim. Yet the same mechanism reveals a deeper concern. If moral attention is directed primarily by emotional salience, then certain forms of suffering may become privileged over others. The image does not merely communicate information; it shapes what observers perceive as morally significant. Consequently, moral influence depends not only on the reality being depicted but also on the psychological reactions elicited by the depiction itself.

3.4 From Moral Reactions to Moral Judgments

This effect of emotion highlights an important distinction between moral reactions and moral judgments. Moral reactions occur prior to conscious reflection, whereas moral judgments emerge as a result of reflection. A moral reaction is therefore an immediate, pre-reflective response to a situation. One may feel disgust, compassion, anger, or admiration before engaging in any explicit moral reasoning. A moral judgment, by contrast, is the result of

evaluating those initial responses alongside one’s broader beliefs, principles, and reasons. Although moral reactions often contribute to moral judgments, the two should not be treated as identical.

This distinction is important because people are capable of reasoning about their reactions. An individual may initially react negatively to a disturbing image yet later conclude that the depicted action was morally justified. Likewise, a person may experience an immediate positive reaction toward a persuasive image before subsequently recognizing that it was manipulative or deceptive. The capacity for reflection allows moral judgments to diverge from initial emotional responses. In this sense, reason functions not as a replacement for moral reactions but as a mechanism for evaluating, revising, and occasionally overriding them.

The distinction is further supported by the phenomenon of confabulation. In neuropsychiatry, confabulation refers to the generation of explanations or memories that are sincerely believed despite not accurately reflecting the processes that produced them [6] [4]. “The patient believes the statement to be truthful, hence the descriptive term ‘honest lying’” [5]. Although often studied in clinical contexts, similar effects can occur in healthy individuals. People frequently provide rationalizations for their beliefs and actions that are more coherent than the actual cognitive processes that generated them. This does not imply that moral reasoning is irrelevant to the subsequent moral judgment. Rather, it suggests that the relationship between reasoning and judgment is more complex than it is often assumed to be. A person’s stated rationale may accurately describe the reasons that support a judgment, but it may fail to describe the *causal* process that originally produced the judgment. I do not deny that moral reasoning plays a significant role in the formation of moral judgments. Nor do I deny that moral reactions often exert a powerful influence on our evaluations. Instead, I argue that moral reactions and moral judgments correspond to distinct stages of cognition. I argue this because neither can be fully reduced to the other.

This distinction has important implications for machine learning systems. Vision-language models in particular are trained on data generated by human beings; however, there is no mechanism analogous to the moral reaction vs. moral judgment consideration when learning on morally salient visual data. Current work is exploring the reasoning capabilities of such

models [10] [41] [35] [12]; however, as I will show in the following chapter, if the model does not have a fairly developed high-level conceptual understanding, the resulting techniques will not generalize to value alignment.

This flaw is because the training data contains a combination of various immediate moral reactions and reflective moral judgments produced by different annotators under different circumstances. This is magnified by the fact that existing datasets do not distinguish between the existing values or beliefs of the annotators. The result is a training signal that often combines incompatible rationales and value structures. Since visual media is particularly effective at eliciting immediate moral reactions, multimodal systems may be especially susceptible to annotation variances that humans would ordinarily moderate through reflection. Understanding the distinction between moral value, moral reactions, moral influence, and moral judgment is therefore essential for evaluating whether VLMs are capable of representing moral concepts in a coherent and consistent manner.

Chapter 4

THE CROSS-MODAL CONSISTENCY OF VLM VALUE JUDGMENTS

4.1 Introduction

As machine learning systems become increasingly integrated into decision-making processes, ensuring that these systems can make reliable moral and value-based judgments has emerged as an important area of research. Many studies have evaluated the moral judgment capabilities of LLMs, but VLMs have been somewhat underexplored by comparison. The current literature that attempts to evaluate or improve VLM performance on moral benchmarks has largely overlooked a fundamental challenge associated with multimodal learning: the *consistency* of moral judgments across modalities. Accuracy is typically the target for systems such as these, with the defense that improving accuracy will improve many other downstream tasks. The issue is that, unlike LLMs, VLMs are typically constructed by aligning visual and textual embedding spaces, enabling information from images and text to be represented within a shared latent space. This alignment has proven effective for tasks requiring low-level semantic understanding, such as image captioning and visual question answering (VQA). However, it remains unclear whether the same alignment mechanisms are sufficient to support higher-level conceptual reasoning.

High-level conceptual understanding differs from low-level semantic understanding in that it requires abstract concepts, values, and relationships to be represented consistently across modalities. Consequently, evaluating moral judgments in VLMs provides an opportunity to assess whether current multimodal architectures achieve the degree of cross-modal alignment necessary for consistent value-based reasoning. If a model interprets morally equivalent scenarios differently depending on whether they are presented as text or images, this would suggest limitations in the shared representation space and raise concerns about the reliability of multimodal value alignment.

This work investigates the extent to which state-of-the-art VLMs maintain cross-modal consistency in their value judgments. I evaluate models by comparing their moral assessments of semantically equivalent scenarios presented in both textual and visual forms and measure the degree of agreement using the concordance correlation coefficient. My results show that few of the evaluated models achieve a high level of cross-modal concordance, and none of them achieve a distributionally robust concordance coefficient, indicating that current VLMs lack robust high-level conceptual consistency across modalities. These findings have important implications for future research on multimodal value alignment, suggesting that improvements in cross-modal embedding space alignment may be a prerequisite for developing VLMs capable of reliable and consistent moral reasoning.

4.2 *Related Work*

4.2.1 *Moral Data*

Recent efforts to improve such capabilities have focused predominantly on the language domain, with researchers developing large corpora of morally salient text to train and fine-tune large language models (LLMs). Datasets such as ETHICS [16], SOCIAL CHEMISTRY [13], MORAL STORIES [12], and SCRUPLES [26] all take this approach by using various approaches to collecting data regarding what people value. Although these datasets differ in annotation methodology and moral framework, they share a common objective: providing structured representations of human values that can be used both for evaluation and for alignment-oriented training. These datasets have been used as both benchmarks and as fine-tuning datasets for language models to improve the human value alignment of models. Systems such as DELPHI [19] [20] take this even further by training with the objective of evaluation against human moral judgments.

Despite this progress, the study of moral alignment in multimodal systems remains comparatively underdeveloped. Until recently, there were few datasets designed to evaluate moral reasoning in contexts involving both visual and textual information. This gap has motivated the development of multimodal benchmarks such as M3oralBench [38], VIVA [18], DIVE [29], and MORALISE [24], which develop image-text datasets with structures

analogous to those of the text-only counterparts. These datasets extend many of the goals of text-only moral benchmarks to vision-language settings by incorporating images alongside textual descriptions, thereby enabling the evaluation of value judgments grounded in visual contexts.

Image-Text Alignment

Modern vision-language models rely on an alignment mechanism that maps visual representations into a form that can be processed by a language model. The most widely adopted approach uses a lightweight projection module, typically a linear layer or a shallow Multi-Layer Perceptron (MLP), to transform visual features extracted by a vision encoder into the hidden representation space of the LLM. This input allows the LLM to process visual information alongside text [3] [11], yet despite its popularity, newer models are exploring alternatives to this standard.

While simple projection-based approaches have proven highly effective, alternative architectures have been proposed to improve cross-modal alignment. One prominent example is the Query Transformer (Q-Former), introduced in models like BLIP-2 [22] and later InstructBLIP [9]. A Q-Former to extract informative features tailored to textual instructions. This approach uses cross-attention to project the image modality to the text, allowing for more complex feature relationships to be captured in the process.

Moral Alignment

Despite these many advancements within the broader space of multimodal machine learning, the focus of the field until recently has been on low-level semantic understanding. Tasks such as VQA or image segmentation focus on training a model that can identify patterns in images and associate those patterns with words that represent concepts. This progression has continued until researchers began to work with high-level conceptual understanding. Since then, the representations of high-level concepts have been of utmost importance through projects such as commonsense reasoning [8]. Rapid developments in moral understanding have been made within NLP, and so researchers have begun investigating whether similar

capabilities extend to vision-language models (VLMs)[10].

Recent work suggests that vision-language models are capable of representing morally relevant information, but significant challenges remain. Existing studies primarily evaluate whether models can identify moral violations or produce socially acceptable responses. Comparatively little attention has been devoted to understanding how value concepts are represented internally, how these representations differ across modalities, or whether textual and visual inputs produce consistent value judgments. As a result, an important gap remains between demonstrating moral competence on benchmark tasks and understanding the underlying semantic mechanisms that support moral reasoning. Addressing this gap is essential for developing multimodal systems whose value judgments remain stable and consistent across different forms of input.

4.3 Methods

4.3.1 Datasets

imSitu

imSitu is a dataset aimed at supporting situation recognition [39]. The goal of the dataset is to present a variety of images, each having a concise description of the situation with special focus on the main activity, other involved agents/objects, and the roles of these agents/objects in the situation. Built on linguistic frames from FrameNet, each image is annotated with linguistically correct attributes by selecting from the set of existing actions and roles that could occur (e.g., riding, cooking, repairing). The dataset contains around 26,000 images with 3 situations per image.

For the context of this work, imSitu will be used as a baseline for samples in which the image and text depict the same situation. The structure of imSitu is important because it does not aim to simply caption, but rather emphasizes the actions and agents involved. This is important because image captions often miss these details, a factor that is often crucial in value judgments, as we have seen in the previous chapters.

COCO

COCO (Common Objects in Context) is a dataset with similar aims as imSitu, and focuses on captioning images for the downstream goal of applying learned patterns in images that will likely be encountered in the real world (hence the “common” aspect) [23]. It also contains roughly 200,000 labeled images, allowing for sufficient variance for the evaluation within this work. The important part is that COCO images are focused on the scene-level, which allows for tasks such as object localization and semantic segmentation. This scene-level focus, combined with the rich text descriptions, allows COCO to serve as a dataset to validate any findings within imSitu.

MORALISE

MORALISE is a benchmark dataset designed to evaluate the moral alignment of vision-language models (VLMs) in realistic multimodal settings [24]. This dataset is especially unique because few works have specifically aimed to achieve multimodal value alignment. The goal of the dataset is to present image-text pairs that contain morally salient content and assess whether models can correctly identify moral violations and reason about the underlying norms. MORALISE organizes moral violations into 13 topics and specifies personal, interpersonal, and societal domains. It is important to note that these labels are not relevant to this paper, since I only aim to assess the cross-modal *consistency* of VLMs, not necessarily their accuracy when compared to human benchmarks of value. MORALISE aims to be used for multiple downstream tasks and provides both image labels and captions for a multitude of applications. The focus on normative aspects of human behavior allows for a deeper exploration of the factors that do (or don’t) influence a VLM’s value judgment.

4.3.2 Value Pairs

As established, the term “value” can encapsulate a wide range of evaluative concepts, but often asserts some form of normativity. When we assert something as *valuable*, we apply some objective, universal nature to the thing in question. As was previously defended, when something is deemed to be valuable, it means that one has good reasons to value that thing.

This distinction is extremely important because what we are beginning to extract is the meaning of axiology. Axiology is also known as value theory, and is the study of evaluative concepts regarding that which is good. Within this work, I primarily focus on axiological ethics, or evaluative concepts regarding that which is right; however, it is important to note that the primary distinction comes from the normativity of ethical/moral concepts. Normativity refers to the standards that people or things ought to, or should, be held to. Normativity is effective to assess not only from a philosophical perspective, but also from a technical perspective.

This effectiveness is rooted in the overwhelming breadth of axiology. Axiology as a field encompasses a wide variety of evaluative concepts, such as beauty, creativity, happiness, or even meaning. While these concepts ought to be explored within deep learning, they are far beyond the scope of the current literature on value representation in machine learning systems. By limiting my focus to normative values, I can create a clear distinction between positive and negative values that can apply to a wide variety of multimodal scenes. This means that both text and image representations of the scenes can be normatively evaluated as *right* or *wrong* (in some way) without relying on some otherwise modal-specific aspect to the judgment (for instance, like how it would be difficult to describe a beautiful image in text). As such, I focus on normative values purely as a testbed to assess current VLM capabilities. It is important to distinguish the fact that *I do not aim to evaluate VLM moral performance*; rather, I purely explore the consistency of existing VLM value judgments. The values selected are shown in Figure 4.1.

It is important to note that every word represents the core concept of *value* in varying levels of specificity. For example, “moral” and “immoral” are the positive and negative terms used to denote what is “right” and “wrong,” respectively. Similarly, the pair of “ethical” and “unethical” *as well as* “permissible” and “impermissible” are typically used to also represent “right” and “wrong.” Finally, “good” and “bad” extend beyond purely “moral/immoral” by representing the normative concept of value (and specifically intrinsic value as discussed in Chapter 1).

Positive Value Concept	Negative Value Concept
moral	immoral
good	bad
right	wrong
ethical	unethical
permissible	impermissible

Table 4.1: The normative value concepts being evaluated.

4.3.3 Models

The models selected for this evaluation have been specifically selected due to their many architectural similarities and differences. In addition, have elected to only assess open-source vision-language models. This decision was made because many of the fundamental issues I will explore are related to the architecture and training data, attributes that are often hidden by teams developing closed-source systems. The models that will be evaluated are LLaVa [25], Qwen [3], InternVL [42], InstructBLIP [9], and Molmo [11]. The exact versions and specifications of the evaluated models are detailed in Tables 4.2 and 4.3.

To contextualize the importance of these architectural similarities, we must examine the core structure behind a generative VLM. Generative VLMs are typically constructed by taking an LLM and an image encoder. One starts by freezing the image encoder (and usually also freezing the text encoder), then training a projection layer to translate the image encoder’s image embeddings to a similar space as the text encoder’s text embeddings. The goal through this process is to make the translated image embeddings isomorphic to the text embeddings for the same concepts. After this alignment process, image tokens are added to the text encoder’s tokenizer. These image tokens are used as a signal during the model’s forward pass to substitute text embeddings for image embeddings (which should be in the same embedding space after the alignment process). Depending on the model, more fine-tuning will occur after this alignment, but the core process is completed after training the projection layer.

Model Name	Text Encoder	Image Encoder	Projection Layer	Pre-Training Data	
llava-v1.5-7b	Vicuna-7B	CLIP ViT-L/14	Linear layer	595K image-text pairs (CC3M); 558K filtered pairs (LAION/CC/SBU, captioned by BLIP)	
llava-v1.5-13b	Vicuna-13B	CLIP ViT-L/14	Linear layer	595K image-text pairs (CC3M)	
instructblip-vicuna-7b	Vicuna-7B	CLIP ViT-g/14	Q-Former	129M images (COCO, Visual Genome, CC3M, CC12M, SBU); 115M images (LAION400M)	
instructblip-vicuna-13b	Vicuna-13B	CLIP ViT-g/14	Q-Former	129M images (COCO, Visual Genome, CC3M, CC12M, SBU); 115M images (LAION400M)	

Table 4.2: Vision-Language Models: Architecture and Training Data (Vicuna-Based Text Encoders)

Model Name	Text Encoder	Image Encoder	Projection Layer	Pre-Training Data
Qwen3-VL-2B-Instruct	Qwen3-2B	SigLIP2-Large (300M)	Two-layer MLP	Not publicized (36T tokens)
Qwen3-VL-4B-Instruct	Qwen3-4B	SigLIP2-Large (300M)	Two-layer MLP	Not publicized (36T tokens)
Qwen3-VL-8B-Instruct	Qwen3-8B	SigLIP2-SO-400M	Two-layer MLP	Not publicized (36T tokens)
Molmo-7B-D	Qwen2-7B	CLIP ViT-L/14	MLP	PixMo-Cap
InternVL3-1B	Qwen2.5-0.5B	InternViT-300M-448px-V2.5	MLP	4.98B samples (LAION-en, LAION-multi, LAION-COCO, etc.)
InternVL3-2B	Qwen2.5-1.5B	InternViT-300M-448px-V2.5	MLP	4.98B samples (LAION-en, LAION-multi, LAION-COCO, etc.)
InternVL3-8B	Qwen2.5-7B	InternViT-300M-448px-V2.5	MLP	4.98B samples (including LAION-en, LAIONmulti, LAION-COCO, etc.)

Table 4.3: Vision-Language Models: Architecture and Training Data (Qwen-Based Text Encoders)

With this understanding, one may note that many models share either the same image or the same text encoders; however, all of the model families vary in their projection strategies. This distinction will become increasingly important as we discuss the implications of the results in the following sections. Another relevant distinction was the fact that only the “instruct” versions of each model were used. This means that they were instruction-tuned, but didn’t fully go through human-preference optimization. This choice was crucial to the validity of the results because the model checkpoint should be able to respond to instructions and completion tasks, but wouldn’t have its outputs augmented by human-preference nor safety alignment that was outside of the core training data. This allows for a clearer picture regarding why the model outputs take a certain form.

4.3.4 *Prompting*

Prompting strategy can significantly affect the quality of model outputs, and can even go so far as to increase the accuracy if the correct strategy is utilized [37]. This is why I leverage a simple prompting scheme. The goal of this evaluation is to assess the VLM’s core concept-level consistency, rather than assessing which prompts allow for variances in consistency. With this in mind, the prompt used for the text-only evaluations is as follows:

$$s_t = t(\text{“[scenario]’ Is this scenario [pos value] or [neg value]? This scenario is”})$$

Where t is the model’s tokenizer. Since we are working with VLMs, there is also a chance that we could be provided an image. The prompt, when providing an image, roughly adheres to the following structure:

$$s_i = t(\text{“[image] Is this scenario [pos value] or [neg value]? This scenario is”})$$

This prompt specifically aims to assess the next token probability, hence the lack of punctuation or other structural templates (such as “User: ... Assistant: ...”). As a result, I can isolate the next token likelihood without the prompt itself serving as a confounding variable.

In addition, I have intentionally included the values to be evaluated within the prompt. This has the effect of reducing the model’s long-form response characteristics and raises

the likelihood of the positive and negative value tokens appearing. Since they are both included, the model’s conditioning allows for an increase in the likelihood of both tokens. This becomes crucial because I only utilize the logits from the model’s forward pass, and no other aspects of its generation paradigm, meaning that I am leveraging the direct conditional probability.

The fact that the values are direct antonyms of one another is also an important distinction. The antonymic nature of the values provided means that they *should* never move in likelihood together for a value judgment, but rather they should move inversely to one another. In other words, when making a judgment that uses the literal words, one value should be high, and the other should be low. This relative nature of “high” and “low” is a problem without the normalized positive value probability as presented in Section 4.3.5.

4.3.5 Evaluation Metrics

Normalized Positive Value Probability

To derive the normalized positive value probability, we must first understand what is meant by the positive value probability. I define the positive value probability as follows:

$$\mathbb{P}(v^+ | s_t) = \prod_{i=1}^{|v^+|} \mathbb{P}(v_i^+ | (s_t, v_{<i}^+))$$

Where v^+ is the list of tokens that come from tokenizing the positive value concept, v_i^+ is the token at index i , s_t is the (ordered) list of tokens from text-only prompt sentence as described above, and $(s_t, v_{<i}^+)$ denotes the concatenation of all tokens from v that are less than i to the end of s_t .

Since we will need to compare the probabilities between the positive and negative concepts, we will want to normalize this positive concept probability. The formulation for the normalized positive concept probability is as follows:

$$\hat{\mathbb{P}}(v^+ | s_t) = \frac{\mathbb{P}(v^+ | s_t)}{\mathbb{P}(v^+ | s_t) + \mathbb{P}(v^- | s_t)}$$

Effectively what is being represented here, is the probability of selecting the positive

value token *over* the negative one. This relationship combined with their antonymic nature is all that is needed to make a value judgment. If we used the raw non-normalized values, it would be unlikely to get a positive value judgment since the probability mass of the resulting distribution over the entire vocabulary would skew the models output to be closer to 0. By normalizing the value, I am emulating the process of top-1 token selection, where the largest token is simply selected. As a result, this normalized positive value probability takes into account the ratio between the positive and negative value tokens, representing the model’s confidence in its prediction as well.

Concordance Correlation Coefficient (Cross-Modal Value Alignment)

To quantify the cross-modal conceptual agreement, I utilize the concordance correlation coefficient (CCC). The formulation for the CCC is as follows:

$$\frac{2\rho\sigma_x\sigma_y}{\sigma_x^2 + \sigma_y^2 + (\mu_x - \mu_y)^2}$$

Where ρ is the Pearson’s correlation coefficient, μ is the mean for a specified variable, and σ is the standard deviation for a variable.

The CCC is effective for this evaluation because it effectively measures whether the agreement between two variables falls along the $x = y$ plane. Therefore, the metric evaluates *both* precision and accuracy by quantifying whether the values change together and whether there is no statistical bias to the values. Within this work, a higher CCC is better.

Jensen-Shannon Divergence

To quantify the difference between two probability distributions in a symmetric and **bounded** manner, I utilize the Jensen-Shannon (JS) divergence. The formulation for the JS divergence is as follows:

$$D_{JS}(P|Q) = \frac{1}{2}D_{KL}(P|M) + \frac{1}{2}D_{KL}(Q|M)$$

where $M = \frac{1}{2}(P + Q)$ is the pointwise mean of the two distributions, and D_{KL} is the Kullback-Leibler divergence. Expanded fully:

$$D_{JS}(P|Q) = \frac{1}{2} \sum_i P(i) \log \left(\frac{P(i)}{M(i)} \right) + \frac{1}{2} \sum_i Q(i) \log \left(\frac{Q(i)}{M(i)} \right)$$

where P and Q are the two probability distributions being compared and i indexes the possible outcomes.

JS divergence is necessary for this evaluation for quite a few reasons. Unlike KL divergence, JS divergence is symmetric, which means that the result does not depend on the distribution that is designated as the reference. Since we will later be exploring the divergence in correctly vs. incorrectly classified samples, symmetry becomes important to ensure we are focusing on distributional divergence rather than solely information loss. Additionally, JS divergence is bounded to be between $[0, 1]$, which means that JS divergence directly interpretable without requiring an external reference point, something that is not possible with KL divergence.

A major issue is that JS divergence measures distributional differences in continuous probability distributions, but my normalized positive value probabilities are structured as discrete distributions. To estimate continuous probability distributions from discrete samples for use in distribution comparison metrics, I utilize Kernel Density Estimation (KDE). KDE essentially constructs a smooth density estimate by placing a kernel function centered at each observed data point and summing the contributions. The formulation for the KDE estimate at a point x is as follows:

$$\hat{f}(x) = \frac{1}{nh} \sum_{i=1}^n K \left(\frac{x - x_i}{h} \right)$$

where n is the number of samples, h is the bandwidth parameter controlling the smoothness of the estimate, x_i are the observed data points, and K is the kernel function. I utilize the Gaussian kernel:

$$K(u) = \frac{1}{\sqrt{2\pi}} \exp \left(-\frac{u^2}{2} \right)$$

KDE makes no parametric assumptions about the shape of the underlying distribution, meaning that it is especially effective for multimodal distributions. Note that for the way

that JS Divergence will be used, lower values are typically better, since we are aiming for alignment.

4.3.6 Maximum Mean Discrepancy

To quantify the difference between two distributions directly from samples without requiring density estimation, I utilize Maximum Mean Discrepancy (MMD). Unlike KDE-based metrics such as JS divergence which first estimate a continuous density from discrete samples and then compare those densities, MMD operates directly on the raw sample sets, making no assumptions about the shape of the underlying distribution. The formulation for MMD is as follows:

$$\text{MMD}^2(P, Q) = \mathbb{E}_{x, x' \sim P}[k(x, x')] - 2\mathbb{E}_{x \sim P, y \sim Q}[k(x, y)] + \mathbb{E}_{y, y' \sim Q}[k(y, y')]$$

where $k(\cdot, \cdot)$ is a kernel function measuring similarity between points. I utilize the Radial Basis Function (RBF) kernel:

$$k(x, y) = \exp\left(-\frac{\|x - y\|^2}{2\sigma^2}\right)$$

where σ is the bandwidth parameter, selected via the median heuristic $\sigma = \text{median}(\|x_i - x_j\|)$ over all pairwise distances in the combined sample set.

MMD is effective for this evaluation because it is sensitive to differences in the full geometry of the sample distributions, including differences in mean, variance, and higher-order moments without requiring a density estimation step that could smooth over fine-grained structural differences. Within the context of this work, a lower MMD is better.

4.3.7 Sampling Strategy

The datasets utilized for the evaluation vary significantly in size, which becomes an issue as I explore the generalizability of my findings. To control for this variance in size, I conduct random uniform sampling over each dataset to select 10,000 samples that will be used for evaluation.

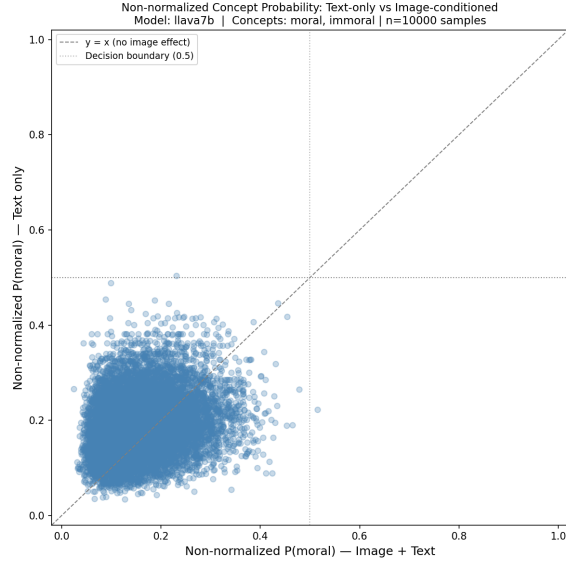


Figure 4.1: The (non-normalized) cross-modal modal probabilities of generating the token “moral” from LLaVa-7B given 10,000 random samples from imSitu

In addition, when sampling images as comparisons for the heatmaps as described in Section 4.4.5. I again conduct uniform random sampling but on the 10,000 samples to get 10 images that will be used to classify the text scenario. These ten images are randomly sampled for every text in the 10,000 samples.

4.4 Results

4.4.1 Cross-Modal Antonymic Probabilities

As stated in Section 4.3.5, the magnitude of the model’s probabilities for the positive and negative value tokens is not guaranteed, which presents an issue regarding the interpretation of VLM value judgments. This is shown clearly in Figure 4.1.

Even though we have alignment along the diagonal representing some form of cross-modal consistency, we cannot interpret these predictions as value judgments. This current plot tells us nothing about how likely it was for the model to choose “moral” *instead of* “immoral,” and it is this distinction that drives the following section.

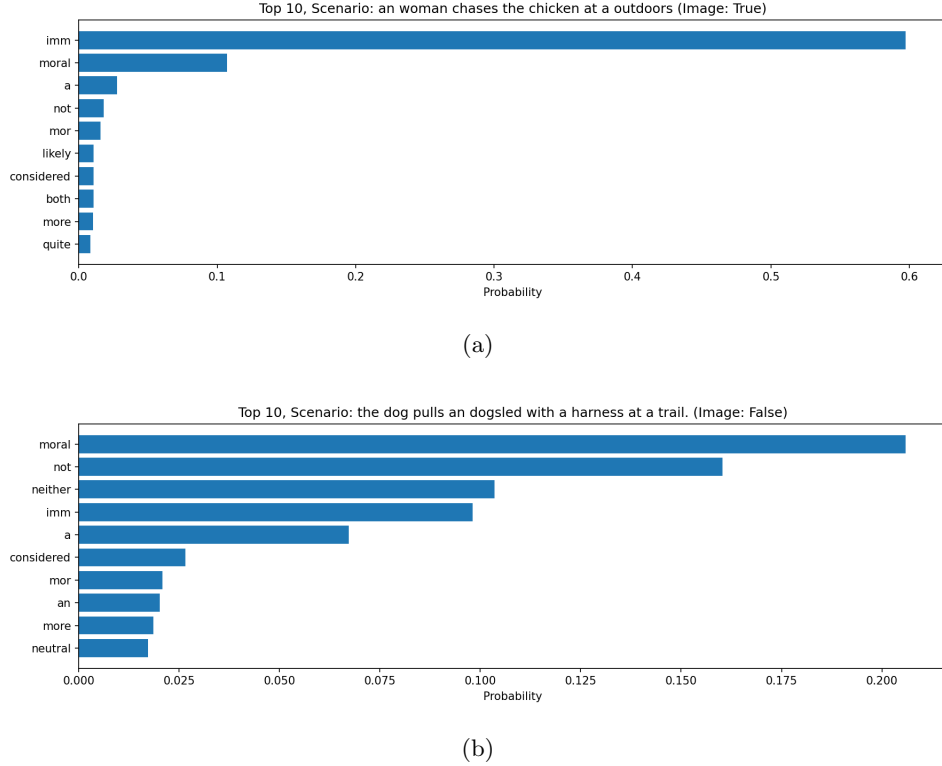


Figure 4.2: Plots of the top 10 tokens for LLaVa-7B when conditioned upon the specified scenario as part of the prompt with value pair being moral/immoral (See Section 4.3.4 for structure)

4.4.2 Normalized Cross-Modal Antonymic Probabilities

By using the normalized positive value probability, we can effectively “force” a selection, then also take into account the relative difference between the pair. Since they are antonyms, if “moral” is to be selected, then “immoral” should have a very low probability, and vice versa. Putting theory into practice, Figure 4.2 demonstrates this understanding. By plotting the top-10 tokens using the specified scenarios, we can see that LLaVa-7B was able to select one of the tokens from the value pair as the predicted token.

This vocabulary distribution has now established that we can safely utilize the normalized positive value probability without issues regarding the probability mass distribution. Now we can look to further analyzing the normalized cross-modal antonymic probabilities.

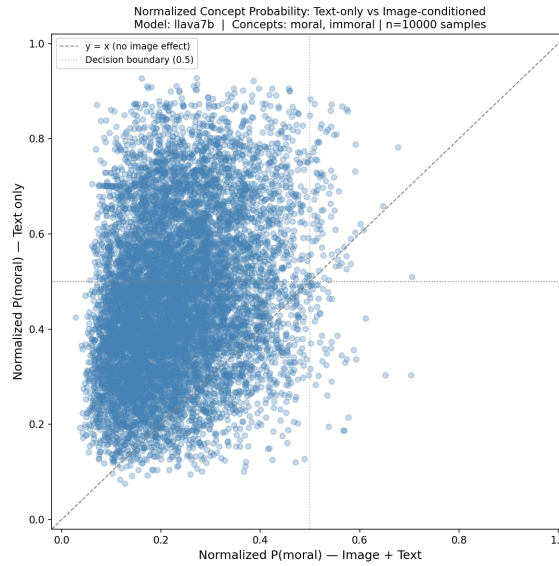


Figure 4.3: The cross-modal modal normalized positive value probability for the value pair moral/immoral from LLaVa-7B on imSitu

Within Figure 4.3 we can finally see the distributional differences that arise.

Now we can clearly see the distributional differences between the tokens for the words *moral* and *immoral*. This means that LLaVa-7B has a heavy skew toward predicting images as immoral, while text is split between the positive and negative tokens.

4.4.3 Cross-Modal Value Consistency

As explored within section 4.3.2, many normative concepts are used in different ways and in different frequencies, despite representing the same fundamental thing. To fully understand the variance that occurs with this lexical shift, we must compare the distributional differences when utilizing alternative words for varying specificity of the same concept. To reiterate, all of the concepts being utilized represent *value*; however, each specifies things to varying degrees of normativity.

Here we can see that, as hypothesized, the distributional difference in these evaluative concepts results in significant differences in the resulting probability mass of the cross-modal normalized positive value probabilities. A few interesting things of note come up

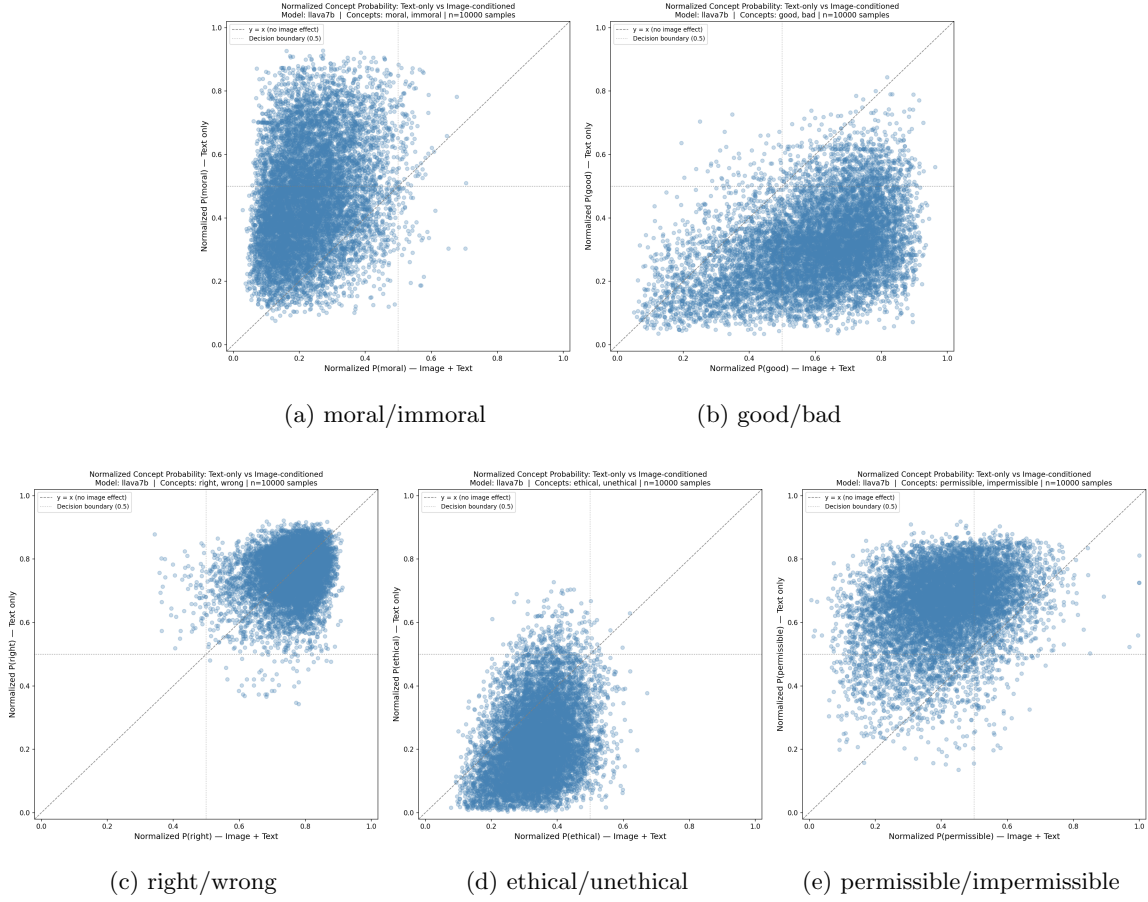


Figure 4.4: Plots of the cross-modal normalized positive value probabilities of LLaVa-7B with the specified value pairs.

upon viewing these plots. (1) All of the plots are somewhat skewed toward the $y = x$ line, which means that although the model isn't perfectly aligned, its predictions are at least somewhat consistent. (2) Certain predictions, like those for the token pair right/wrong, are quite cross-modally consistent, yet tend to cluster in their specific evaluation results. This means that this model may not be effectively utilizing the context provided from the situations, and as a result, its predictions stay consistent no matter the scenario given. The cross-modal consistency of the model can be quantified by the CCC, and is shown in Table 4.4. This metric validates what was mentioned before. As we can see, the value pair right and wrong produces the plot that is considered to be the most aligned, yet all of the plots

have some skew toward $y = x$. The CCC does not correspond to clustering, which is why the right/wrong token pair was able to achieve such a high score.

Value Pair	CCC
moral/immoral	0.096
right/wrong	0.231
ethical/unethical	0.191
permissible/impermissible	0.100
good/bad	0.137

Table 4.4: The LLaVa-7B CCC across all value pairs for imSitu

4.4.4 Inter-Model Cross-Modal Value Consistency

Although we have now seen how the normalized cross-modal can vary significantly across values, it is still unclear whether this trend holds across all VLMs. To better understand this variation, Figure 4.5 plots all five models on the same data to showcase the differences.

Here, it becomes clear that most models fail to maintain any semblance of cross-modal consistency, despite being of a similar order of magnitude in size to LLaVa-7B. A glance will tell you that the only other model that has a meaningful trend toward/along the $y = x$ line is InternVL-8B. To quantify this relationship, we can once again utilize the CCC as is shown in Table 4.5.

4.4.5 The Effect of Visual Semantic Understanding on Cross-Modal Value Consistency

The poor cross-modal consistency of the majority of the evaluated models directly leads to the following question: Do the models not understand what the images represent? If a model cannot reliably identify the actions, agents, objects, or relationships depicted in an image, then inconsistencies between text-only and image-text evaluations could be a consequence of weak visual understanding.

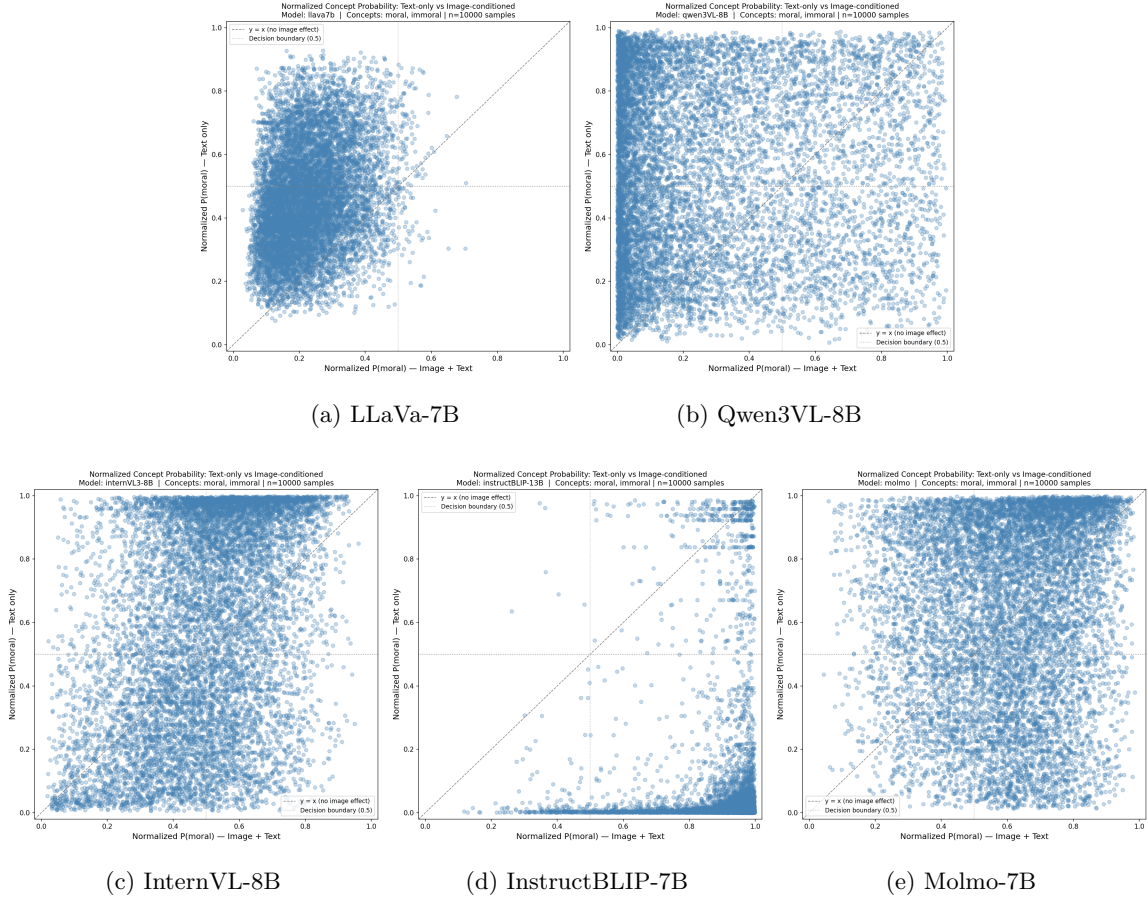


Figure 4.5: Plots of the cross-modal normalized positive value probability of the specified models on the value pair moral/immoral.

This distinction is important because the primary objective of this work is to evaluate value alignment and value consistency, not general image recognition performance. A model that incorrectly identifies a situation depicted in an image cannot reasonably be expected to produce value judgments that align with those derived from a textual description of the same situation. Conversely, if a model demonstrates strong visual semantic understanding while still exhibiting inconsistent value judgments across modalities, this would provide stronger evidence that the inconsistency originates from the value reasoning process itself.

To investigate this possibility, the models are evaluated on their ability to recognize and interpret the semantic content of images. The evaluation is conducted at two levels of

Model	CCC
llava7b	0.096
llava13b	0.299
qwen3VL-2B	0.067
qwen3VL-4B	0.022
qwen3VL-8B	0.030
internVL3-1B	0.159
internVL3-2B	0.206
internVL3-8B	0.307
instructBLIP-7B	0.230
instructBLIP-13B	-0.003
molmo	0.095

Table 4.5: The moral/immoral CCC across all models for imSitu

semantic abstraction:

1. Image captioning performance is examined using COCO, which primarily measures scene-level understanding, object recognition, and the ability to generate coherent descriptions of visual content.
2. Situation recognition performance is evaluated using imSitu, which places greater emphasis on identifying actions, agents, and semantic roles within an event.

To answer this question, I begin by developing heatmaps that represent the probability of selecting a scenario from a list of scenarios, given an image. This heatmap serves as a form of image classification, where the texts that are correctly identified as describing their specified image will go on to be labeled as correctly-identified samples. A sample heatmap for LLaVa-7B is given in Figure 4.6.

Please note that 4.6 only shows a small subset of the data for interpretability; however,

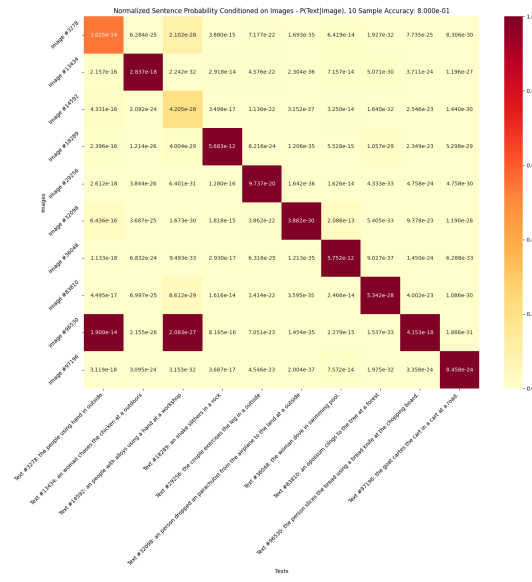


Figure 4.6: A heatmap representing the probability of generating the target text given the defined image using LLaVa-7B. The ground truth falls along the diagonal.

the full evaluation sampled 10,000 texts and also sampled 10 random images for every text in order to form the full heatmap. By this, we can now see that our models do not fully understand every image, so let’s examine the distributional differences in correctly classified vs. incorrectly classified samples as shown in Figure 4.7.

Once again, a cursory glance will show that there is no significant divergence in distribution from the correctly vs. the incorrectly classified images. Despite this, still need to understand whether this behavior extends across all other value pairs. Figure ?? shows just this, and we can see that the incorrectly classified samples seem to track with the probability mass of the correctly classified samples, so we can look to Table 4.6 to interpret the metrics.

This being said, Figure 4.5 clearly showed that LLaVa was one of the best performing models, so to fully understand the impact of visual semantics on the resulting judgments, we must examine how robust each the model is to distributional variance by modifying the dataset being used.

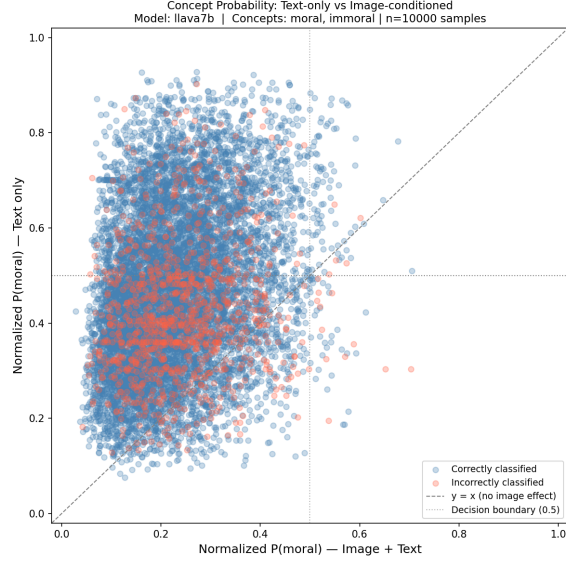


Figure 4.7: The cross-modal normalized positive value probability of LLaVa-7B for the token pair moral/immoral. Incorrectly classified samples are colored orange.

The Robustness of Cross-Modal Value Consistency

The analyses thus far have focused primarily on the cross-modal alignment, first exploring how different value pairs are represented within the model, then moving on to exploring how different models represent the same concept. While these measures give insight into the overall level of cross-modal consistency, they do not examine whether the observed relationships are stable across datasets with different semantic characteristics. For example, a model may exhibit reasonable consistency on one dataset that is closer to its original training data, while failing to maintain the same behavior when the underlying visual content changes.

To assess this robustness, I utilize three metrics: the CCC to evaluate cross-modal agreement, JS Divergence to measure the similarity between underlying distributions, and MMD to have a non-parametric estimate of the distance between the distributions of the correctly and incorrectly classified samples. The results are shown in Table ??.

Several observations can be made. First, the absolute CCC values remain relatively low across all evaluated datasets, indicating that even when textual and multimodal representa-

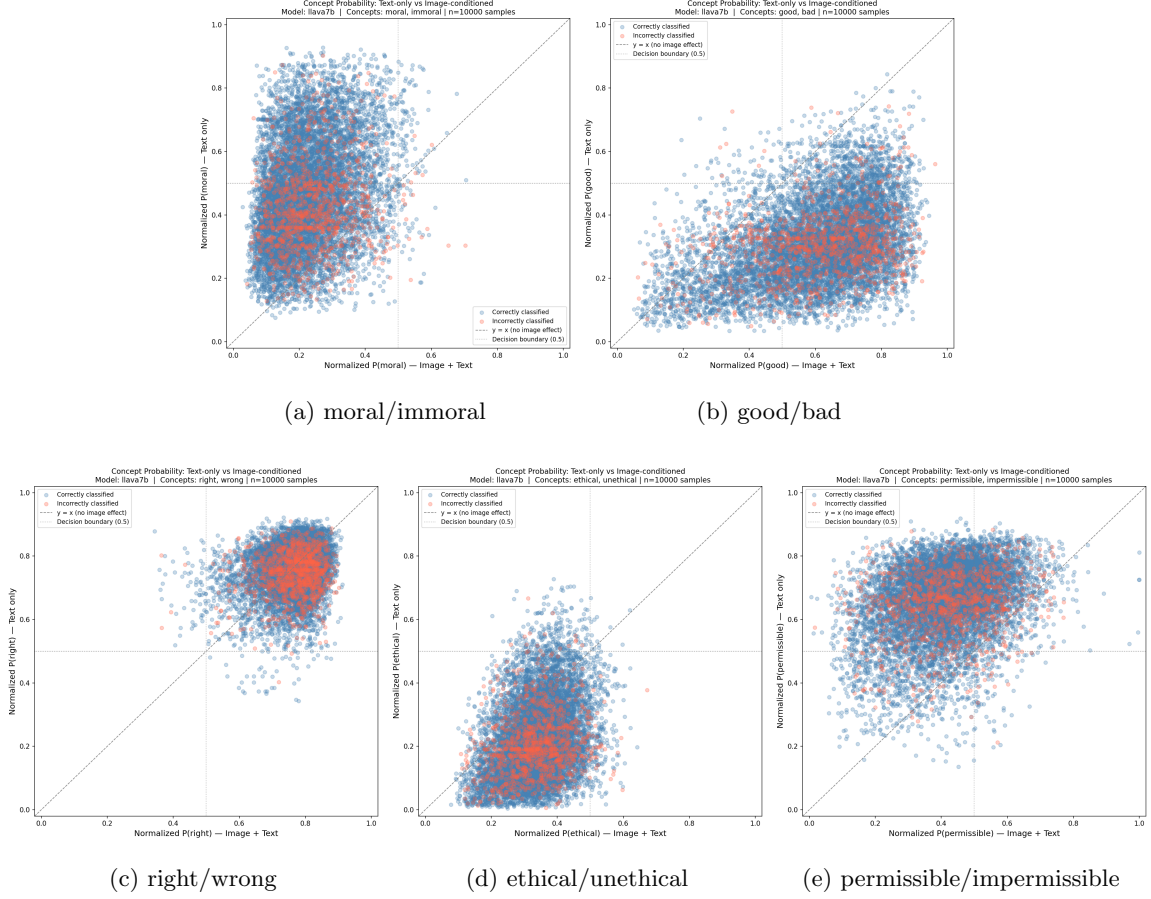


Figure 4.8: Plots of the image classification results in the cross-modal normalized positive value probability of LLaVa-7B. Each subplot uses the specified value pair.

tions are related, the strength of their agreement remains limited. This finding is consistent with the cross-modal inconsistency observed throughout the previous experiments. Second, the relatively small JS divergence values suggest that the distributions themselves are not drastically different, even when their instance-level agreement is weak. In other words, the models often produce value representations that occupy similar regions of the latent space while still failing to maintain strong correspondence between individual textual and visual samples.

Comparing datasets reveals that consistency is not uniform across semantic levels. The highest CCC values are observed on imSitu, particularly for LLaVA-13B, suggesting that

Concept Pair	CCC	JS Divergence	MMD
moral/immoral	0.096	0.025	0.020
right/wrong	0.231	0.018	0.011
ethical/unethical	0.191	0.017	0.009
permissible/impermissible	0.100	0.013	0.004
good/bad	0.137	0.020	0.004
moral/immoral	0.145	0.017	0.007

Table 4.6: The distributional divergence of the correctly and incorrectly classified images from LLaVa-7B (classification accuracy=0.893).

Model	Dataset	CCC	JS Divergence	MMD
LLaVa-7B	imSitu	0.096	0.025	0.020
LLaVa-7B	COCO	0.020	0.013	0.001
LLaVa-7B	MORALISE	0.070	0.016	0.007
LLaVa-13B	imSitu	0.299	0.020	0.021
LLaVa-13B	MORALISE	0.155	0.016	0.009

Table 4.7: The distributional divergence of LLaVa models across varying datasets

structured situation descriptions involving explicit actions and semantic roles provide a stronger basis for aligning visual and textual value representations. By contrast, COCO exhibits the lowest agreement despite also producing low divergence scores. This result suggests that scene-level descriptions alone may be insufficient for establishing robust value correspondence, as they frequently omit the actions, intentions, and social relationships that are most relevant for value judgments.

The comparison between LLaVA-7B and LLaVA-13B further indicates that increasing model scale improves robustness. The larger model consistently achieves higher CCC scores while maintaining similar divergence values, implying that additional capacity primarily

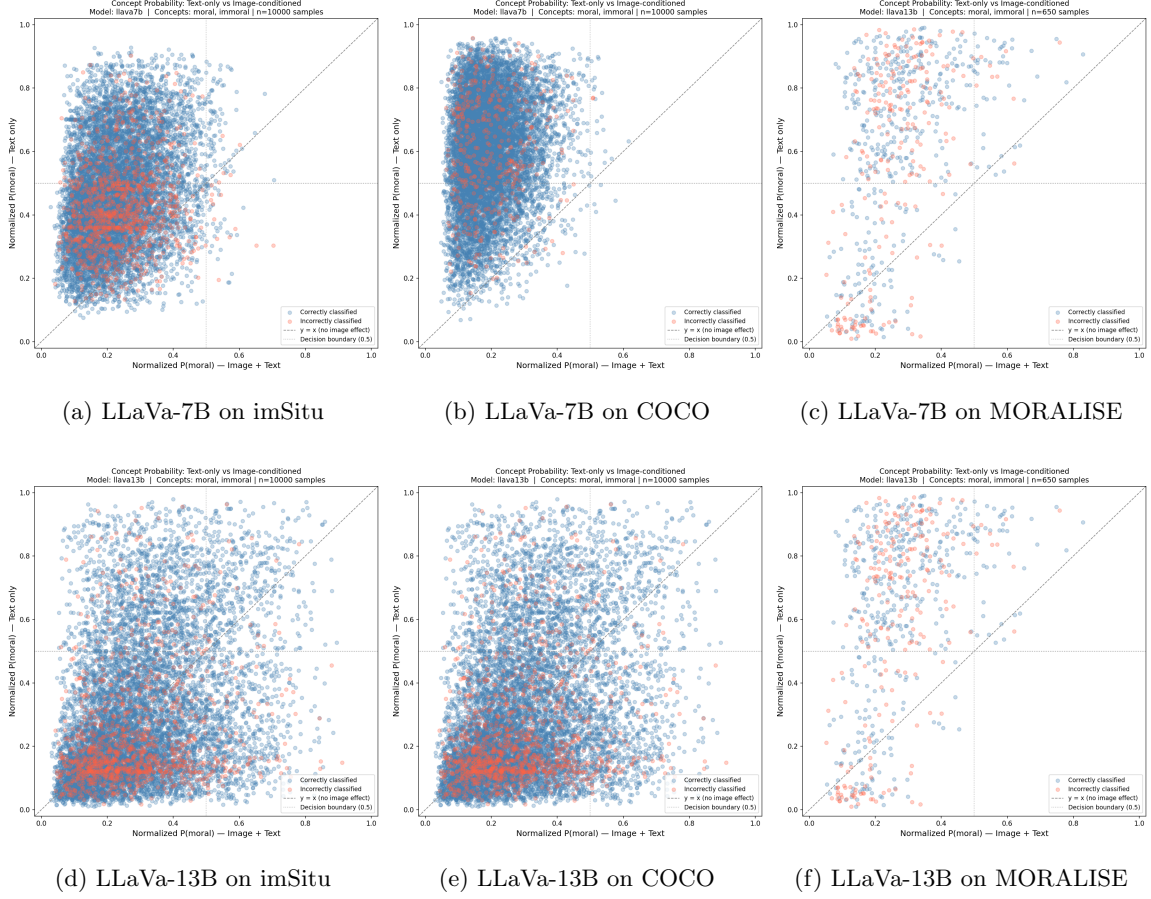


Figure 4.9: Plots of the image classification results in the cross-modal normalized positive value probability of LLaVa-7B. Each subplot uses the specified value pair.

strengthens the correspondence between modalities rather than simply reshaping the overall distribution. Nevertheless, the improvements remain modest, and substantial disagreement persists even in the strongest-performing configurations.

Taken together, these results suggest that cross-modal value consistency is relatively robust at the distributional level but considerably weaker at the instance level. While models tend to learn broadly similar value distributions across modalities, they often struggle to consistently align individual visual situations with their corresponding textual value judgments. This finding reinforces the conclusion that value reasoning represents a distinct challenge beyond general semantic understanding, requiring not only accurate recognition

of situations but also stable mappings between those situations and higher-level value concepts.

4.4.6 *The Effect of Scale on High-Level Semantic Understanding*

One possible explanation for the poor cross-modal value consistency observed throughout the previous experiments is insufficient model capacity. In many other benchmarks, larger models have repeatedly been shown to exhibit improved performance on a wide range of reasoning, instruction-following, and multimodal understanding tasks. If value judgments emerge from higher-level semantic representations, then increasing the scale of a vision-language model may improve its ability to consistently associate visual situations with corresponding value concepts.

To investigate this hypothesis, Figure 4.10 presents the cross-modal normalized positive value probability plots for model families at multiple parameter scales using the value pair moral/immoral. Because all models within a family share a common architecture and training methodology, differences between scales provide insight into whether additional parameters improve the alignment between visual and textual value representations.

Several quite unintuitive results become apparent. (1) Even though increasing scale generally results in more structured value distributions, the inverse appears here. Smaller models exhibit tightly packed clusters, and as model size increases, clusters become less concentrated. This pattern indicates that larger models may be losing some semantic information, or have a less-refined projection layer training phase when compared to their smaller counterparts. This would fit what is being shown, where if the projection layer stays consistent across model sizes, then the larger models should require more time and more data to adequately converge.

This being said, there are still improvements in the performance of both the LLaVa and InternVL families of models. This is notable, because they were the only scaled models I evaluated that focused on simplicity in the projection layer. This simplicity would allow the model to learn without a large amount of data or training time, and in fact, the authors of InternVL stated "With the same number of parameters, the depth, head

dimension, and MLP ratio have little impact on the performance“ [7]. This shows that the relationship between scale and value understanding is not linear. While larger models tend to demonstrate deeper image understanding and reasoning capabilities, this suggests that increasing parameter count alone is insufficient to fully resolve the challenge of cross-modal value reasoning.

This effect is particularly interesting on the problem of value consistency (and even more broadly high-level cross-modal semantic consistency). If value consistency were solely a consequence of improved visual semantics, one would expect a correspondingly large increase in the separation of moral and immoral samples as scale increases. The persistence of overlap in Figure 4.10 suggests that value reasoning introduces an additional layer of complexity beyond situation recognition alone. Models may correctly identify the actions and agents depicted in an image while still failing to consistently map those situations onto stable value representations. Taken together, these results indicate that while model scale often contributes positively to low-level semantic understanding, but that scaling alone does not guarantee cross-modal value consistency. The findings therefore support the view that moral and value-based reasoning constitute a distinct challenge that extends beyond general visual-semantic competence.

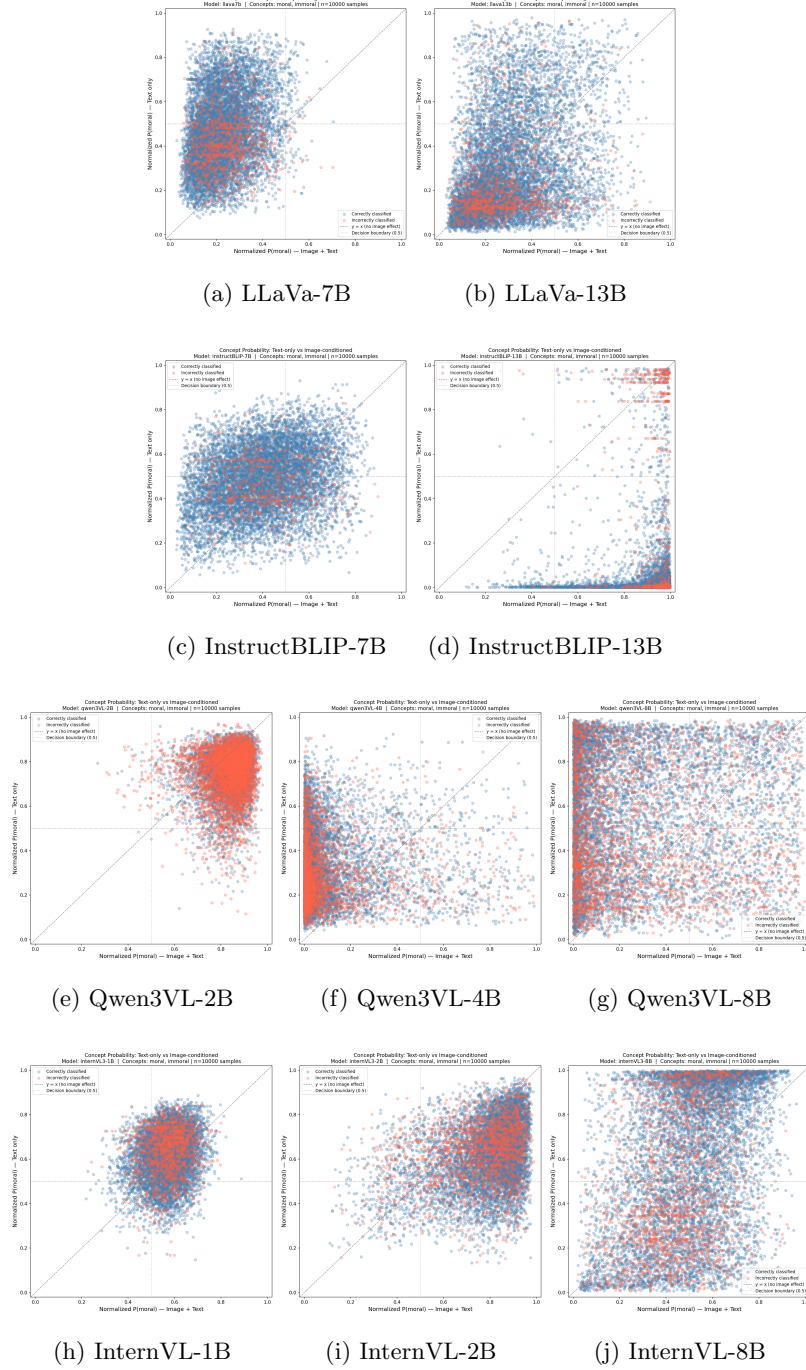


Figure 4.10: Cross-modal normalized positive value probability plots in order of increasing scale for each model, using the value pair *moral/immoral*.

Model	Dataset	CCC	JS Divergence	MMD
llava7b	moralise	0.070	0.016	0.007
llava7b	imsitu	0.096	0.025	0.020
llava7b	coco	0.020	0.013	0.001
llava13b	moralise	0.155	0.016	0.009
llava13b	imsitu	0.299	0.020	0.021
qwen3VL-2B	imsitu	0.067	0.004	0.001
qwen3VL-2B	coco	-0.004	0.016	0.018
qwen3VL-4B	imsitu	0.022	0.007	0.004
qwen3VL-8B	imsitu	0.030	0.011	0.012
internVL3-1B	imsitu	0.159	0.010	0.005
internVL3-2B	imsitu	0.206	0.015	0.009
internVL3-8B	imsitu	0.307	0.027	0.037
instructBLIP-7B	moralise	0.477	0.071	0.126
instructBLIP-7B	imsitu	0.230	0.019	0.008
instructBLIP-7B	coco	0.243	0.037	0.016
instructBLIP-13B	moralise	-0.001	0.207	0.029
instructBLIP-13B	imsitu	-0.003	0.238	0.079
molmo	imsitu	0.095	0.029	0.038

Table 4.8: All model moral/immoral distribution differences across datasets

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